Search Smooths Discontinuities in Discrete/Continuous Problems

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Abstract

When used to solve models with both discrete and continuous choices, the Endogenous Gridpoint Method (EGM), in addition to finding the optimal solution, can also generate points in the solution that are suboptimal. This paper shows that incorporating search into such problems can smooth out the discontinuities in the decision rules, which are responsible for producing the suboptimal points. With ample search frictions, the suboptimal points can be eliminated entirely. In addition to providing greater control over the smoothness of the problem, search offers the added benefit of rationalizing the smoothness into the agent's choice problem.

Keywords: Consumption, Computational Economics, Discrete Continuous Choice, Endogenous Gridpoint Method, Search
JEL codes: C13, C63

Repo: https://github.com/AMonninger/DCEGM_Search
PDF: https://github.com/AMonninger/DCEGM_Search/Text/dcegmSearch.pdf

The computational results in this paper were constructed using tools in the Econ-ARK/HARK toolkit. The toolkit can be cited by its digital object identifier 10.5281/zenodo.1001068, as is done in the paper's own references as **?**. Thanks to Matthew White for initial discussions of this subject.

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1 Introduction

This paper introduces new methods for efficiently solving dynamic optimization problems with both discrete and continuous choices (DC models). These methods extend the Endogenous Gridpoint Method (EGM) [Carroll, 2006] by including exogenous outcome probabilities, search frictions, and taste shocks to 'concavify' the value function of the optimization problem. Compared to existing extensions of the EGM for DC models, the methods introduced in this paper have the added advantage of not only providing greater smoothness, but also rationalizing the smoothness into the agent's choice problem.

For dynamic stochastic optimization problems with continuous choice variables, the Endogenous Gridpoint Method (EGM) is significantly more efficient compared to standard root-finding methods in finding the optimal decision rules. In these problems, the value functions are typically concave and the Euler equation is a necessary and sufficient condition for the optimal decision rule. In contrast, when the choice variables include both continuous and discrete choices (DC models), the Euler equation has multiple solutions for the continuous choice. Therefore, the Euler equation is only a necessary, but not sufficient, condition for the optimal decision rule.

The complication in solving a dynamic optimization problem arises due to the kinks in the value function and discontinuities in the decision rule at the points in the state space where discrete choices are made. These discontinuities make the decision rule non-monotonic, which leads to the Euler equation producing suboptimal points. In addition, each period's kinks and discontinuities are propagated back in time as the solution is solved by backward iteration, thus, exacerbating the problem. Fella [2014], Druedahl and Jørgensen [2017], and Iskhakov et al. [2017] provide methods for finding and discarding these suboptimal points from the final solution.

In order to mitigate the complications arising from the accumulation of the discontinuities, Iskhakov et al. [2017] introduce Extreme Value Type I taste shocks that affect the likelihood of the discrete choice outcomes. These taste shocks have the advantage of smoothing out the expected value function and the expected marginal utility function. With larger taste shocks, the value function can be 'concavified' to a greater degree. Such taste shocks, however, are in fact 'behavioral' shocks that add a degree of randomization to the discrete choice outcomes.

In contrast, this paper introduces search frictions whereby agents exert search effort that determines the likelihood of changing the discrete state. In addition, it includes log-normally distributed taste shocks that affect the utility that agents derive from making discrete choices. Combined, these two features not only lead to greater smoothness of the decision rule, they also rationalize the smoothness arising from the taste shocks, as these shocks determine the search effort that the agents exert.

To illustrate the properties of search as a smoothing mechanism and to compare it to other modeling features that can also provide smoothness, we use a standard consumption-saving model in which the agent can also receive the 'option' to make a binary retirement decision, depending on the search effort that they make. If the agent makes no search effort, they receive the option to retire with exogenous probability p and if they make the maximum effort, they receive the option to retire with certainty. Effort, however, is costly and the cost function is assumed to be a convex function of effort. In addition, households receive taste shocks that affect the relative value they derive from working and are also exposed to income uncertainty.

All four of the aforementioned modeling features, i.e. search, exogenous switching probability, taste shocks, and income shocks, serve to smooth out the decision rule in the DC model with varying degrees of effectiveness. While search provides the highest degree of smoothness, in terms of reducing the size of the discontinuities in the decision rule, it has to be combined with taste shocks to yield uniform smoothness around the discontinuities. Moreover, each of the smoothing mecha-

nisms have different economic interpretations and affect the decision rules differently. While greater smoothness from search and exogenous switching probability shifts the consumption function upwards, greater smoothness from taste shocks and income shocks shifts the consumption function downwards. Separately, we show that a coarse income-shock discretization introduces *phantom kinks*—spurious numerical non-monotonicities that should not be confused with the economic discontinuities from discrete retirement—and that integrating the continuous income density with a transition-matrix method removes those artifacts at moderate computational cost (Section 2.5).

2 Retirement Model

In this section, we build a consumption-saving model and extend it to include the option to retire and search frictions. Agents in this model derive utility from consuming non- durable goods c_t and get disutility from working. The disutility parameter is denoted by δ. Each period, there is a possibility for the agents to receive an 'option' to retire. The likelihood of receiving the option depends on the amount of search effort ε_{t-1} that the agent exerted in the previous period. Search effort, however, is costly and we assume that the search cost is a convex function of effort, with a functional form given by $\theta(\varepsilon_t) = \sigma_s\left(\varepsilon_t + (1-\varepsilon_t)\log(1-\varepsilon_t)\right)$. This cost function implies that the initial marginal effort is cheap; however, greater effort that increases the likelihood of receiving the option to switch the discrete state leads to increasingly larger costs. Assuming logarithmic utility from consumption, and denoting the choice to retire and the choice to work by $d_t = 0$ and $d_t = 1$, respectively, the utility function for the agent is given by

$$u(c_t, d_t, \varepsilon_t) = \log(c_t) - \delta d_t - \theta(\varepsilon_t). \tag{1}$$

At the end of each period, households can save their end-of-period liquid assets A_t at a risk free rate R. Working in the current period determines whether the agent receives income y_t in the following period. Therefore, the beginning-of-period assets are given by $M_t = RA_t + y_t d_{t-1}$. Furthermore, we assume that $y_t = y\eta_t$, where η_t follows a mean-one log-normal distribution, $\eta_t \sim \mathcal{N}(-\sigma_\eta^2/2, \sigma_\eta^2)$.

2.1 Retiree's sub-problem

Retirement is assumed to be self-absorbing, so the problem reduces to the simple consumption-saving problem, which in Bellman form is given by:

$$V_t^{\text{Ret}}(M_t) = \max_{0\le c_t\le M_t} \left\{u(c_t, 0, 0) + \beta V_{t+1}^{\text{Ret}}(M_{t+1})\right\}, \tag{2}$$

where $M_{t+1} = A_t R$ and the end-of-period assets $A_t = M_t - c_t$. Using the gothic script V ($\mathcal{V}$) to represent the value of ending the period with assets A_t, the end-of-period value for a retired agent is, by definition

$$\mathcal{V}_t^{\text{Ret}}(A_t) \equiv V_{t+1}^{\text{Ret}}(M_{t+1}(A_t)) = V_{t+1}^{\text{Ret}}(A_t R). \tag{3}$$

Thus, the problem in terms of the end-of-period assets can be restated as

$$\mathcal{V}_{t-1}^{\text{Ret}}(A_{t-1}) = \max_{0\le c_t\le A_{t-1}R} \left\{\log(c_t) + \beta \mathcal{V}_t^{\text{Ret}}(A_t)\right\}. \tag{4}$$

2.2 Worker's sub-problem

Each period, a worker either receives the option to retire or has to continue working. There is an exogenous probability p of receiving the option to retire. In addition, a worker can increase the

likelihood of receiving the option to retire by exerting search effort. The search effort is made at the end of a period and affects the likelihood of receiving the option to retire in the following period. When period t starts, an income shock is realized. This is followed by the realization of a taste shock, ξ_t, which is a multiplicative shock that affects the utility that a worker derives from working. The taste shocks are assumed to follow a mean-one log-normal distribution, i.e. $\xi_t \sim \mathcal{N}(-\sigma_\xi^2/2, \sigma_\xi^2)$. These taste shocks are different from the Extreme Value Type I shocks in Iskhakov et al. [2017] in two major ways. Firstly, the assumption of log-normal taste shocks means that these shocks can take up multiple distinct values in the discrete approximation of the distribution. Secondly, these shocks are not the purely behavioral shocks that randomize the discrete choice outcome. Instead, these taste shocks affect the likelihood of the discrete outcome by determining the search effort that the agent exerts.

Based on the search effort made in the previous period, if the agent receives the option to switch, the agent decides whether to retire or continue working. If the agent does not receive the option to switch, then they continue to work. Having made the discrete choice, the agent makes a consumption decision. If the agent chooses to retire, they stay as a retiree, because the assumption that retirement is self-absorbing. If the agent chooses to continue working, they exert search effort, which then determines the likelihood of receiving the option to switch in the next period. Fig. 1 outlines the timeline of the realization of various shocks and the decisions in the worker's problem.

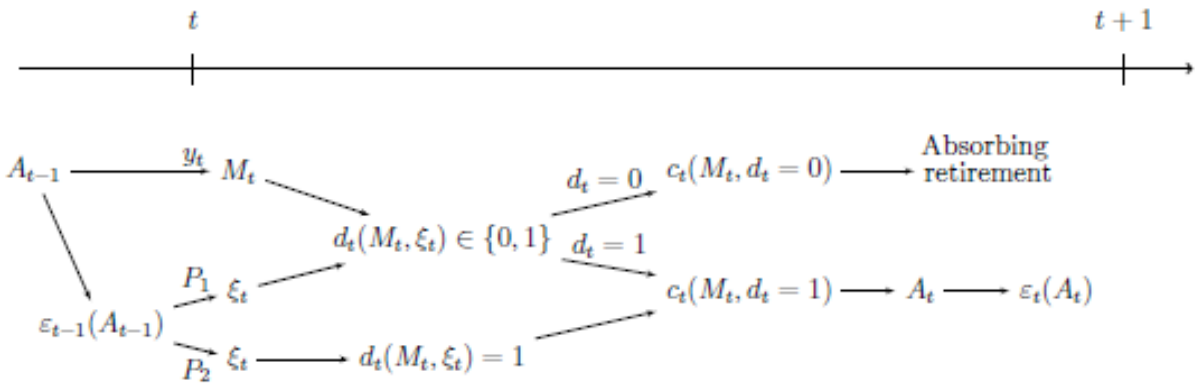


Figure 1: Timeline of the worker's problem

Let the set of possible choice sets for the worker's discrete choice be given by $\mathcal{D}_t \in \{\{\text{Retire}, \text{Work}\}, \{\text{Work}\}\} \equiv \{\{d_t = 0, d_t = 1\}, \{d_t = 1\}\}$. The first element of this set is the case when the agent has the option to switch and the second element is the case when the agent does not have the option to switch and must continue working. For the case when $\mathcal{D}_t = \{W\}$, i.e. the worker does not have the option to switch, the Bellman equation is given by:

$$V_t(M_t, \{W\}, \xi_t) = \xi_t \cdot \max_{\substack{0 \le c_t \le M_t \\ 0 \le \varepsilon_t \le 1}} \left\{ u(c_t, 1, \varepsilon_t) + \beta P_1(\varepsilon_t) \mathcal{V}_t^1(A_t) + \beta P_2(\varepsilon_t) \mathcal{V}_t^2(A_t) \right\} \tag{5}$$

where $\mathcal{V}_t^1(A_t)$ is the expected value of saving A_t when having the option to retire in $t+1$, $P_1(\varepsilon_t) = (1-\varepsilon_t)p + \varepsilon_t$ is the probability of receiving the option to retire in $t+1$, given the search effort ε_t in the current period, $\mathcal{V}_t^2(A_t)$ is the expected value of saving A_t when not having the option to retire in $t+1$, and $P_2(\varepsilon_t) = (1-\varepsilon_t)(1-p)$ is the probability of not receiving the option to retire in $t+1$, given the search effort ε_t in the current period.

For the case when $\mathcal{D}_t = \{R, W\}$, i.e. the case when the worker has the option to switch, the

Bellman equation is given by:

$$V_t(M_t, \{R, W\}, \xi_t) = \max_{d_t \in \{0,1\}} \left[V_t^{\text{Ret}}(M_t);\ V_t(M_t, \{W\}, \xi_t) \right], \tag{6}$$

or

$$\begin{aligned} V_t(M_t, \{R, W\}, \xi_t) = \max_{d_t \in \{0,1\}} \Bigg[& \max_{0 \le c_t \le M_t} \left\{ u(c_t, 0, 0) + \beta \mathcal{V}_t^{\text{Ret}}(A_t) \right\}; \\ & \xi_t \cdot \max_{\substack{0 \le c_t \le M_t \\ 0 \le \varepsilon_t \le 1}} \left\{ u(c_t, 1, \varepsilon_t) + \beta P_1(\varepsilon_t) \mathcal{V}_t^1(A_t) + \beta P_2(\varepsilon_t) \mathcal{V}_t^2(A_t) \right\} \Bigg]. \end{aligned} \tag{7}$$

The expected value of saving A_t when having the option to retire in $t+1$, i.e. $\mathcal{V}_t^1(A_t)$, is given by

$$\mathcal{V}_t^1(A_t) = \int_{y_{t+1}} \int_{\xi_{t+1}} V_{t+1}(M_{t+1}(A_t), \{R, W\}, \xi_{t+1})\, dF(y_{t+1})\, dF(\xi_{t+1}), \tag{8}$$

where $M_{t+1} = A_t R + d_t y$. Similarly, the expected value of saving A_t when not having the option to retire in $t+1$, i.e. $\mathcal{V}_t^2(A_t)$, is given by

$$\mathcal{V}_t^2(A_t) = \int_{y_{t+1}} \int_{\xi_{t+1}} V_{t+1}(M_{t+1}(A_t), \{W\}, \xi_{t+1})\, dF(y_{t+1})\, dF(\xi_{t+1}). \tag{9}$$

2.3 Optimal effort and consumption functions

Since we assume that the consumption decision is made before the effort decision, (5) can be split into two sequential sub-period problems and rewritten as:

$$V_t(M_t, \{W\}, \xi_t) = \xi_t \max_{0 \le c_t \le M_t} \left[\mathcal{V}_t(A_t) \right] \tag{10}$$

where

$$\mathcal{V}_t(A_t) = \max_{0 \le \varepsilon_t \le 1} \left\{ u(c_t, 1, \varepsilon_t) + \beta P_1(\varepsilon_t) \mathcal{V}_t^1(A_t) + \beta P_2(\varepsilon_t) \mathcal{V}_t^2(A_t) \right\} \tag{11}$$

Thus, the first order condition of (11) gives the optimal effort as a function of the end-of-period assets that are determined after the optimal consumption decision has been made in the previous sub-period. The optimal effort is given by

$$\frac{\partial \theta(\varepsilon_t)}{\partial \varepsilon_t} = \beta \mathcal{V}_t^1(A_t) \frac{\partial P_1(\varepsilon_t)}{\partial \varepsilon_t} + \beta \mathcal{V}_t^2(A_t) \frac{\partial P_2(\varepsilon_t)}{\partial \varepsilon_t} \tag{12}$$

$$-\sigma_s \log(1 - \varepsilon_t^*) = \beta(1-p) \mathcal{V}_t^1(A_t) - \beta(1-p) \mathcal{V}_t^2(A_t) \tag{13}$$

$$\varepsilon_t^* = 1 - \exp\left(-\frac{\beta(1-p)}{\sigma_s} \left(\mathcal{V}_t^1(A_t) - \mathcal{V}_t^2(A_t) \right) \right). \tag{14}$$

This expression states that optimal effort is an increasing function of the excess value of receiving the option to switch in the next period over not receiving the option to switch.

Next, we derive the expressions for the optimal consumption functions for the retiree and the worker. Since retirement is self-absorbing, the retiree's consumption function is relatively straightforward to derive and is given by:

$$u'(c_t, 0, 0) = R\beta u'(c_{t+1}, 0, 0) \tag{15}$$

or

$$c_t^{\text{Ret}} = \left(R\beta \left(c_{t+1}^{\text{Ret}}(A_t R) \right)^{-1} \right)^{-1}. \tag{16}$$

To derive the consumption function for the worker, we start with the solution to the problem of a worker who does not have the option to retire, i.e. $\mathcal{D}_t = \{W\}$. The first-order-condition for problem (10), with respect to c_t, is given by

$$u'(c_t, 1, \varepsilon_t^*) = \beta \left[P_1(\varepsilon_t^*) \frac{\partial \mathcal{V}_t^1(A_t)}{\partial A_t} + P_2(\varepsilon_t^*) \frac{\partial \mathcal{V}_t^2(A_t)}{\partial A_t} \right], \tag{17}$$

where ε_t^* is the optimal effort function, which is determined in (14), and the derivatives of the end-of-period values are given by

$$\frac{\partial \mathcal{V}_t^1(A_t)}{\partial A_t} = \mathbb{E}_t \left\{ \frac{\partial V_{t+1}^{R,W}(M_{t+1}, \xi_{t+1})}{\partial M_{t+1}} \frac{\partial M_{t+1}}{\partial A_t} \right\} = R\, \mathbb{E}_t \left\{ \frac{\partial V_{t+1}^{R,W}(M_{t+1}, \xi_{t+1})}{\partial M_{t+1}} \right\} \tag{18}$$

and

$$\frac{\partial \mathcal{V}_t^2(A_t)}{\partial A_t} = R\, \mathbb{E}_t \left\{ \frac{\partial V_{t+1}^{W}(M_{t+1}, \xi_{t+1})}{\partial M_{t+1}} \right\}, \tag{19}$$

where the expectation is over next period income and taste shocks, and $V_{t+1}^W(M_{t+1}, \xi_{t+1})$ and $V_{t+1}^{R,W}(M_{t+1}, \xi_{t+1})$, are shorthands for $V_{t+1}(M_{t+1}, \{W\}, \xi_{t+1})$ and $V_{t+1}(M_{t+1}, \{R, W\}, \xi_{t+1})$, respectively. Substituting these expressions into (17) yields

$$u'(c_t, 1, \varepsilon_t^*) = R\beta \left[P_1(\varepsilon_t^*) \mathbb{E}_t \left\{ \frac{\partial V_{t+1}^{R,W}(M_{t+1}, \xi_{t+1})}{\partial M_{t+1}} \right\} + P_2(\varepsilon_t^*) \mathbb{E}_t \left\{ \frac{\partial V_{t+1}^{W}(M_{t+1}, \xi_{t+1})}{\partial M_{t+1}} \right\} \right]. \tag{20}$$

The Envelope Condition for (10) is given by

$$\frac{\partial V_t^W(M_t, \xi_t)}{\partial M_t} = \xi_t \beta \left[P_1(\varepsilon_t^*) \frac{\partial \mathcal{V}_t^1(A_t)}{\partial A_t} + P_2(\varepsilon_t^*) \frac{\partial \mathcal{V}_t^2(A_t)}{\partial A_t} \right] \tag{21}$$

or

$$\frac{\partial V_t^W(M_t, \xi_t)}{\partial M_t} = \xi_t u'(c_t, 1, \varepsilon_t^*) \tag{22}$$

which can be iterated forward to yield

$$\frac{\partial V_{t+1}^W(M_{t+1}, \xi_{t+1})}{\partial M_{t+1}} = \xi_{t+1} u'(c_{t+1}, 1, \varepsilon_{t+1}^*). \tag{23}$$

To derive a similar expression for the derivative of $V_{t+1}^{R,W}$, we note that for the worker who has the option to retire, i.e. $\mathcal{D}_t = \{R, W\}$, the value function is simply the maximum of the value function of the retiree and the value function of the worker who continues to work

$$V_t^{R,W}(M_t, \xi_t) = \max \left\{ V_t^{\text{Ret}}(M_t);\ V_t^W(M_t, \xi_t) \right\}. \tag{24}$$

Therefore, it can be shown that

$$\frac{\partial V_{t+1}^{R,W}(M_{t+1}, \xi_{t+1})}{\partial M_{t+1}} = \mathbb{1}_{t+1}^0 \cdot u'(c_{t+1}, 0, 0) + \mathbb{1}_{t+1}^1 \cdot \xi_{t+1} u'(c_{t+1}, 1, \varepsilon_{t+1}^*), \tag{25}$$

where $\mathbb{1}^0_{t+1} \equiv \mathbb{1}^0_{t+1}(M_{t+1}, \xi_{t+1}) = 1$ if $V^{\text{Ret}}_{t+1}(M_{t+1}) \geq V^W_{t+1}(M_{t+1}, \xi_{t+1})$ and 0 otherwise, and $\mathbb{1}^1_{t+1} \equiv \mathbb{1}^1_{t+1}(M_{t+1}, \xi_{t+1}) = 1$ if $V^{\text{Ret}}_{t+1}(M_{t+1}) < V^W_{t+1}(M_{t+1}, \xi_{t+1})$ and 0 otherwise. Substituting (23) and (25) into (20) yields the consumption function for the worker who continues to work

$$u'(c_t, 1, \varepsilon^*_t) = R\beta\Big[P_1(\varepsilon^*_t)\int_{y_{t+1}}\int_{\xi_{t+1}}\left[\mathbb{1}^0_{t+1}\cdot u'(c_{t+1}, 0, 0) + \mathbb{1}^1_{t+1}\cdot\xi_{t+1}u'(c_{t+1}, 1, \varepsilon^*_{t+1})\right]dF(y_{t+1})\,dF(\xi_{t+1}) \\ + P_2(\varepsilon^*_t)\int_{y_{t+1}}\int_{\xi_{t+1}}\left[\xi_{t+1}u'(c_{t+1}, 1, \varepsilon^*_{t+1})\right]dF(y_{t+1})\,dF(\xi_{t+1})\Big] \tag{26}$$

or

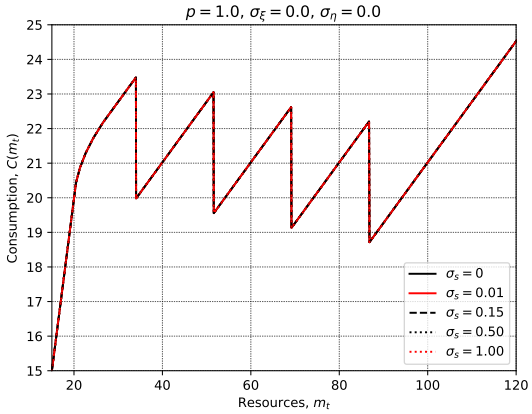
$$c^W_t = \Bigg(R\beta\Big[P_1(\varepsilon^*_t)\int_{y_{t+1}}\int_{\xi_{t+1}}\left[\mathbb{1}^0_{t+1}\left(c^{\text{Ret}}_{t+1}(A_tR)\right)^{-1} + \mathbb{1}^1_{t+1}\xi_{t+1}\left(c^W_{t+1}(A_tR + y_{t+1})\right)^{-1}\right]dF(y_{t+1})\,dF(\xi_{t+1}) \\ + P_2(\varepsilon^*_t)\int_{y_{t+1}}\int_{\xi_{t+1}}\left[\xi_{t+1}\left(c^W_{t+1}(A_tR + y_{t+1})\right)^{-1}\right]dF(y_{t+1})\,dF(\xi_{t+1})\Big]\Bigg)^{-1}. \tag{27}$$

2.4 Results

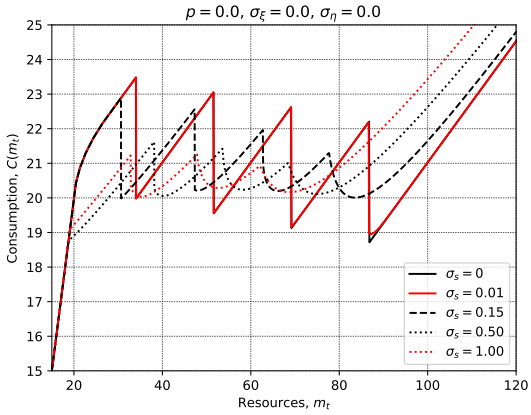
Having established the solution of the retirement problem, we demonstrate how varying the different smoothness parameters affects the solution of the problem. Fig. 2(a) shows the optimal consumption rule of the worker who continues to work in period $T-5$, for the case in which $p = 1.0$. The plot shows the consumption rule for a set of search cost scales, in the absence of income uncertainty and taste shocks. In this scenario, there is no smoothing and the discontinuities that arise from discrete choices being made in periods $T-4$ through $T-1$ are distinctly visible. This is because when $p = 1.0$, the worker receives the option to switch with certainty. In such a case the worker does not exert any search effort, regardless of the search frictions. Consequently, in the case with $p = 1.0$, for all levels of search frictions, we get the same optimal consumption rule, without any smoothing.

In contrast, when $p = 0.0$ (Fig. 2(b)), the worker who does not exert search effort, continues to stay as a worker with certainty. This gives the worker a motive to exert search effort, which determines the likelihood of receiving the option to switch. For a given level of search frictions, decreasing p increases the smoothness of the solution. Moreover, for a given level of p, increasing the search frictions (higher value of σ_s) also increases smoothness. Fig. 2(b) also shows the asymmetric smoothness offered by search, when there are no taste or income shocks. The smoothness happens to the right of the discontinuities. This is because for low liquid resources, workers optimally choose to continue working and hence do not exert any search effort. Workers only exert positive effort to receive the option to retire when they have ample liquid resources. Therefore, the smoothness from search frictions only happens to the right of the discontinuities.

To address this asymmetric smoothness, we need to introduce taste shocks. Since the taste shocks can take a range of different values, there are certain realizations of the taste shock that make the worker want to have the option to switch, even at levels of liquid assets at which, in the absence of a taste shock, the worker would not want to switch. In other words, with the introduction of taste shocks, the range of liquid assets around the discontinuities over which the worker exerts search effort expands. Fig. 3(a) shows that with the incorporation of taste shocks, the consumption function now smoothes out more uniformly around the discontinuities. Moreover, introducing taste shocks allows the search frictions to significantly smooths out most of the discontinuities in the optimal consumption function. This holds even for a very small smoothness parameter for the taste



(a) Have switching option with certainty, $p = 1$.



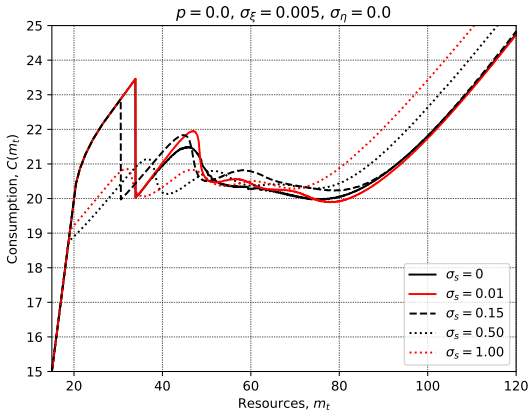
(b) Stay as worker with certainty without search effort, $p = 0$.

Figure 2: The plots show optimal consumption rules of the workers who decides to continue working in period $t = T - 5$ for a set of search cost scales σ_s in the absence of income uncertainty or taste shocks, for the case with $p = 1.0$ (left panel) with $p = 0.0$ (right panel). The rest of the model parameters are $R = 1$, $\beta = 0.98$, $y = 20$.

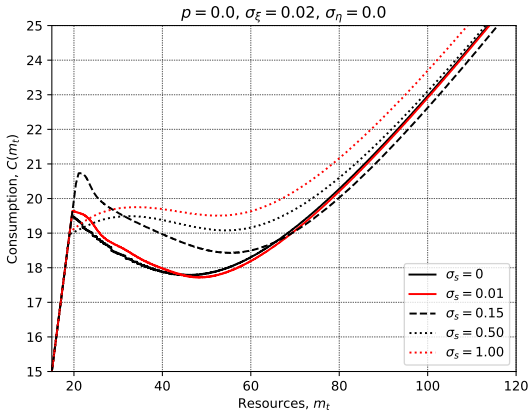
shocks, $\sigma_\xi = 0.005$. This is because now there is a positive likelihood of a worker receiving a taste shock that makes them exert search effort even for values of M_t for which the worker would not exert any search effort in the absence of taste shocks. Increasing the magnitude of the taste shocks to $\sigma_\xi = 0.02$ (Fig. 3(b)) significantly reduces the size of all the discontinuities in the consumption function. Moreover, only moderately sized search frictions are needed to virtually completely smooth out the consumption function.

The results discussed so far do not include any income uncertainty. Adding income uncertainty even into standard consumption-saving models adds curvature and smoothness to the consumption rules. We study the impact of incorporating income uncertainty to the retirement problem on the smoothness of the decision rules in Figs. 4(a) and 4(b). These figures show that while income uncertainty adds smoothness, it does not play a significant role in reducing the size of the discontinuities. Therefore, search frictions and taste shocks play the dominant role in smoothing out the problem. An appropriate combination of search frictions, taste shocks, and income uncertainty can yield a monotonically increasing outcome-specific consumption function by smoothing out the discontinuities that propagate through each period while iterating backwards.

It is important to note that although each of these smoothness mechanisms serve to smooth out the solution for the worker's problem, they have different economic interpretations and affect the optimal consumption rule differently. While lowering the exogenous probability of receiving the option to switch, p, and increasing the search frictions, σ_s, shift the consumption function upwards, large taste shocks, σ_ξ, and income shocks, σ_η, shift the consumption function downwards. Ceteris paribus, decreasing p or increasing σ_s, reduce the likelihood of the worker switching their discrete state successfully. This means that the likelihood of the worker retiring in a given period decreases. As a result, the expected number of years that an agent works increases and they save less in a given period. In contrast, when the size of the taste shocks is increased, the uncertainty surrounding the likelihood of a worker's retirement decision increases. Consequently, the agents save more in a given period, because unexpectedly retiring early would limit the resources that the agent can consume for the remainder of their life. Similarly, when the size of the income shocks increases, households

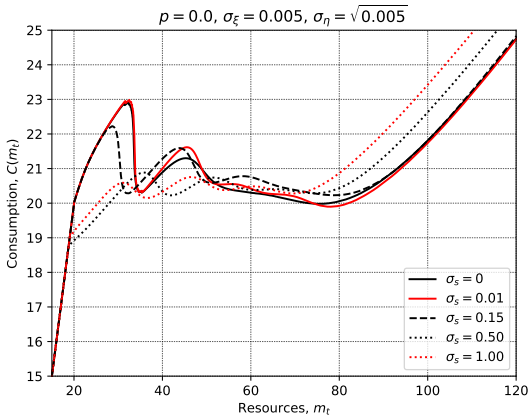


(a) Small taste shocks, $\sigma_\xi = 0.005$.

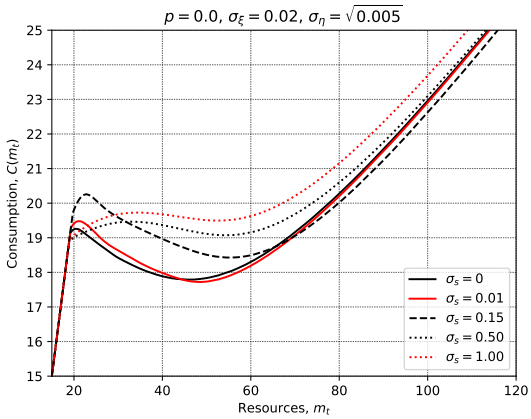


(b) Large taste shocks, $\sigma_\xi = 0.02$.

Figure 3: The plots show optimal consumption rules of the workers who decides to continue working in period $t = T - 5$ for the case with $p = 0$, in the presence of small taste shocks (left panel) and large taste shocks (right panel), without income uncertainty.



(a) Small taste shocks, $\sigma_\xi = 0.005$.

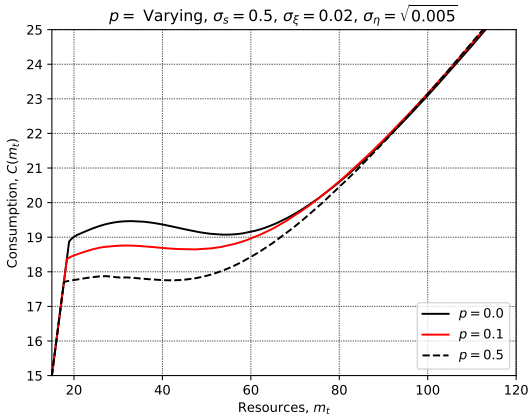


(b) Large taste shocks, $\sigma_\xi = 0.02$.

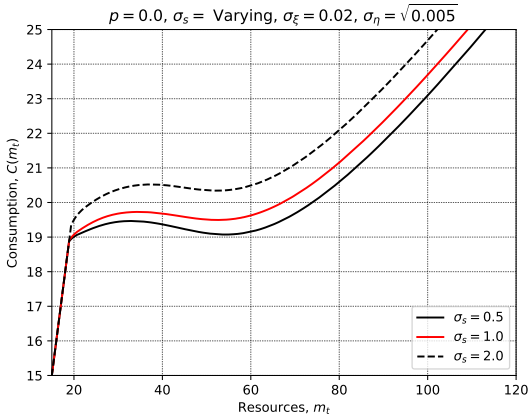
Figure 4: The plots show optimal consumption rules of the workers who decides to continue working in period $t = T - 5$ for the case with $p = 0$, in the presence of small taste shocks (left panel) and large taste shocks (right panel), with income uncertainty.

increase their buffer stock savings, and the consumption function shifts down. Therefore, although each of the smoothness mechanisms achieve the common goal of 'concavifying' the solution, they have different interpretations and different impacts on the optimal consumption rule.

To illustrate these points graphically, we start from the baseline consumption function that has the discontinuities completely smoothed out using smoothness parameters $p = 0$, $\sigma_\xi = 0.02$, $\sigma_s = 0.5$, and $\sigma_\eta = \sqrt{0.005}$. Fig. 5(a) shows the impact of lowering p from the baseline level to $p = 0.1$ and $p = 0.5$, which decreases smoothness, and Fig. 5(b) shows the impact of increasing the search frictions from the baseline level to $\sigma_s = 1.0$ and $\sigma_s = 2.0$, which increases smoothness. As mentioned above, increasing smoothness by adjusting either of the two mechanisms shifts the optimal consumption rule upwards. On the other hand, Fig. 6(a) shows the impact of increasing the size of the taste shocks from the baseline level to $\sigma_\xi = 0.025$ and $\sigma_\xi = 0.03$, and Fig. 6(b) shows the impact of increasing the size of the income shocks from the baseline level to $\sigma_\eta = 5\sqrt{0.005}$ and $\sigma_\eta = 7.5\sqrt{0.005}$. These figures demonstrate that, as expected, increasing the magnitude of the taste or income shocks shifts the optimal consumption function downwards. Therefore, while a combination of these smoothing mechanisms will smooth out the discontinuities, it is also important to note how each of them shift the decision rule differently.



(a) Varying the exogenous probability of receiving the option to switch.

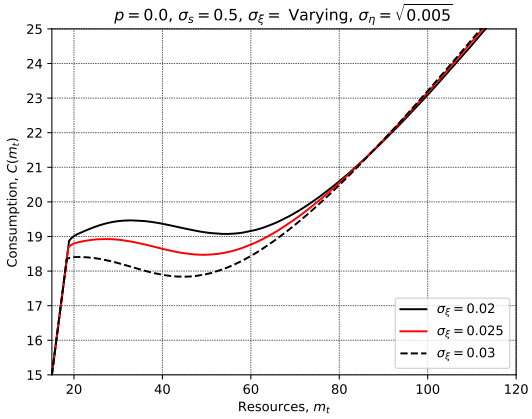


(b) Varying search friction intensity.

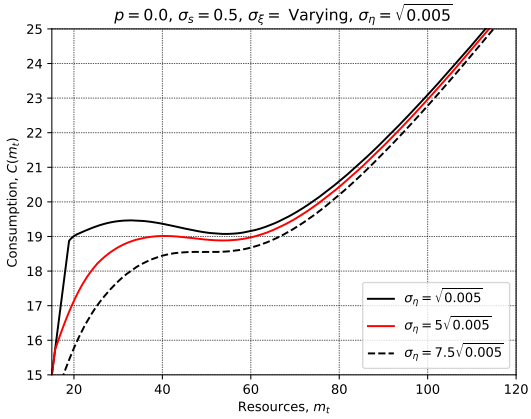
Figure 5: The plots show the impact of increasing the smoothness parameters for the exogenous option probability (left panel) and increasing the magnitude of the search frictions (right panel).

2.5 Income discretization and phantom kinks

The smoothing mechanisms studied above are *economic*: they reshape the agent's decision rule. This subsection instead isolates a *numerical* source of non-smoothness that must not be confused with the economic discontinuities from discrete retirement. When the continuous transitory income density is replaced by a coarse equiprobable grid, the expectation over next-period marginal utility is a sum over a few nodes. That discrete approximation injects *phantom kinks* into the outcome-specific consumption function—spurious non-monotonicities that have no counterpart in the underlying continuous-income problem. The economic kinks associated with the retirement option remain part of the true solution; the phantom kinks should not.



(a) Varying the magnitude of the taste shocks.



(b) Varying the magnitude of the income shocks.

Figure 6: The plots show the impact of increasing the smoothness parameters for the taste shocks (left panel) and increasing the magnitude of the income shocks (right panel).

Experiment. To isolate income integration, we shut down other smoothing channels: $p = 1$ (so $\sigma_s = 0$) and no taste shocks ($\sigma_\xi = 0$, one taste node). We set $\sigma_\eta = \sqrt{0.005}$, matching the paper's income-shock figures, and compare:

1. *Discretization* of the mean-one lognormal transitory shock with $N \in \{1, 7, 31, 151\}$ equiprobable nodes;

2. the *transition-matrix* method (Section 2.6.1), which integrates the same continuous density via a precomputed $b \to m$ matrix (500×510 income grids here).

Other parameters are $R = 1$, $\beta = 0.98$, $y = 20$. Figure 7 plots the working-state consumption rule over $m \in [15, 120]$.

What the figure shows. With $N = 1$ the consumption rule exhibits large sawtooth phantom kinks. Raising N to 7 and then 31 damps those artifacts; at $N = 151$ they are essentially gone and the rule is smooth aside from the economic discrete-choice structure. The transition-matrix solution overlays that fine-discretization *smoothness*: it does not reintroduce the coarse-N zigzags. That is the sense in which the method "works"—it targets the continuous income integral that the discrete nodes only approximate.

Benchmarking caveat. A natural quantitative check is the pointwise gap between each method and the $N = 151$ discretization. We report solve times and that gap in Table 1, but with an important caveat: $N = 151$ is still a *discrete* income approximation, not the continuous-income truth. Small residual differences between the transition matrix and $N = 151$ therefore need not mean the matrix is "wrong"; they can reflect two different approximations to the same continuous expectation. Moreover, those gaps are *horizon-dependent*: near the terminal period the two methods are very close, while earlier periods accumulate integration differences, so a max-absolute deviation versus $N = 151$ can grow even when neither rule displays phantom kinks. The right visual benchmark for the paper's claim is therefore Figure 7 (presence or absence of spurious non-monotonicities), not a claim that the matrix is uniformly closer to $N = 151$ than every coarser N at every t.

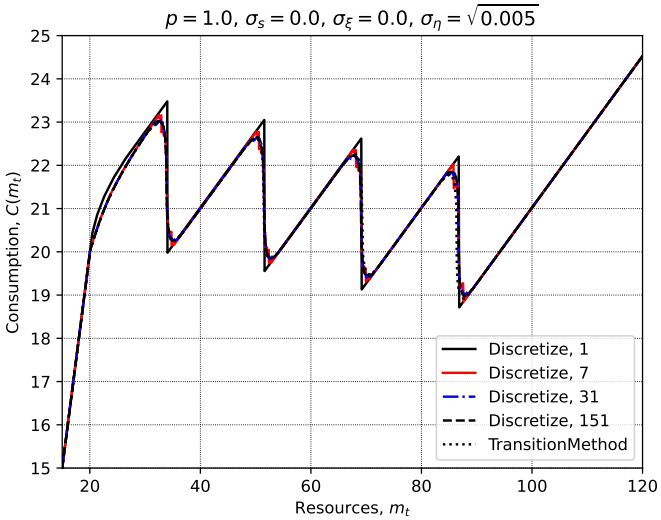


Figure 7: Phantom kinks from income discretization. Coarse N produces spurious wiggles in $C(m)$; refining N (or using the transition matrix) removes them. Held fixed: $\sigma_\eta = \sqrt{0.005}$, $p = 1$, $\sigma_s = 0$, $\sigma_\xi = 0$. The remaining curvature is the economically relevant discrete-choice pattern, not a discretization artifact.

Table 1: Solve time by income-integration method ($\sigma_\eta = \sqrt{0.005}$, $p = 1$, $\sigma_s = 0$, $\sigma_\xi = 0$). Max. / mean abs. deviation of $C(m)$ vs. $N = 151$ on $m \in [15, 120]$ are reported as a diagnostic only (see text); they are not the success criterion for removing phantom kinks.

Method	Gridpoints	Solve time (s)	Max. abs. vs 151	Mean abs. vs 151
Discretization	1	0.50	4.72	0.023
Discretization	7	0.66	4.68	0.004
Discretization	31	1.60	0.094	0.001
Discretization	151	14.3	0	0
Transition matrix	—	4.10	0.034	0.0005

Table 1 also makes the cost point: obtaining a kink-free rule by pushing discretization to $N = 151$ is expensive relative to a moderate transition-matrix grid that achieves the same qualitative smoothness in Figure 7.

Note on shock size. The illustration uses $\sigma_\eta = \sqrt{0.005}$, as in the paper's other income-shock figures. The transition matrix employs a fixed (b, m) support; if σ_η is scaled up substantially (e.g. $2\sqrt{0.005}$), more mass sits near the grid edge and the same matrix can diverge more from a fine discretization unless the income grids are widened and refined (at cost scaling with the product of the grid sizes). The phantom-kink lesson—coarse N creates numerical wiggles; continuous integration or fine N removes them—is not tied to that particular σ_η, but quantitative gaps versus a fixed $N = 151$ benchmark are.

2.6 Algorithm

This section outlines the algorithm for solving the problem of the worker who continues to work in the current period. The problem for the retiree can be pre-solved relatively easily. With the solution of the worker who continues to work and the retiree at hand, it is straightforward to compute the solution of the worker with the option to switch. With ample smoothness, the EGM-step should not produce any suboptimal points. However, if there is not enough smoothness, then an additional 'upper envelope' step is required to remove the suboptimal points. The upper envelope refinement step in Iskhakov et al. [2017] is an example of an algorithm that can be employed at the end of the EGM-step to achieve this.

2.6.1 Transition-matrix income integration

In Algorithm 1, the expectation over the transitory income shock is taken by looping over the J quadrature nodes $\vec{\eta}$ and interpolating the next-period functions at each implied $M^j_{t+1}(A_i)$. As an alternative, the transitory-income integration can be recast as a single (precomputed) linear map, following the transition-matrix approach of White [2015].

Idea. Let $b = RA_t$ denote next-period wealth *before* the income draw, and $m = RA_t + y\eta$ next-period market resources *after* it. For a fixed income process, the conditional density of m given b is time-invariant, so the mapping from a function defined on an m-grid to its conditional expectation on a b-grid can be encoded once as a matrix Π with entries $\Pi_{b,m} \approx \Pr(m \mid b)$.

Steps.

1. Build dense grids over b and m (exponentially spaced) and construct Π from the mean-one lognormal transitory density. Because the m-grid is non-uniform, each density evaluation is weighted by the local grid spacing Δm before row-normalization (so that Π approximates $\int f(m)\, g(m - b)\, dm$ rather than an equally weighted sum over columns). Probability mass mapped outside the tabulated m-range is dropped and each row is renormalized to sum to one.

2. For each next-period quantity of interest (marginal utility and value), evaluate it on the m-grid, then obtain its conditional expectation on the b-grid by the matrix product $\Pi\, f(\vec{m})$. Expectations are then interpolated onto $b = R\,\vec{A}_t$ in inverse form (CRRA-inverse for marginal utilities; $V_T = -1/V$ for values) before the EGM step.

3. The remaining EGM and envelope steps of Algorithm 1 are unchanged.

Algorithm 1: EGM-step with search for the worker who continues to work

Let $\vec{\eta} = \{\eta_1, \ldots, \eta_J\}$ and $\vec{\xi} = \{\xi_1, \ldots, \xi_K\}$ be vectors of quadrature points with associated weights $\vec{\omega} = \{\omega_1, \ldots, \omega_J\}$ and $\vec{\mu} = \{\mu_1, \ldots, \mu_K\}$, respectively.

Form an ascending grid over end-of-period wealth, $\vec{A}_t = \{A_t^1, \ldots, A_t^I\}$

for $i = 1, \ldots, I$ **do**

  for $j = 1, \ldots, J$ **do**

    Compute $M_{t+1}^j(A_i) = RA_i + y\eta_{t+1}^j$

    for $d_{t+1} = 0, 1$ **do**

      Compute $c_{t+1}^{\text{Ret}}(M_{t+1}^j(A_i))$ by interpolating $c_{t+1}^{\text{Ret}}(\vec{M}_{t+1})$ at the point $M_{t+1}^j(A_i)$

      Compute $c_{t+1}^{W}(M_{t+1}^j(A_i))$ by interpolating $c_{t+1}^{W}(\vec{M}_{t+1})$ at the point $M_{t+1}^j(A_i)$

      Compute $V_{t+1}^{\text{Ret}}(M_{t+1}^j(A_i))$ by interpolating $V_{t+1}^{\text{Ret}}(\vec{M}_{t+1})$ at the point $M_{t+1}^j(A_i)$

      for $k = 1, \ldots, K$ **do**

        Compute $V_{t+1}^{W}(M_{t+1}^j(A_i), \xi_k)$ by interpolating $V_{t+1}^{W}(\vec{M}_{t+1})$ at the point $M_{t+1}^j(A_i)$ and multiplying it by ξ_k

        Compute $V_{t+1}^{RW}(M_{t+1}^j(A_i), \xi_k) = \max\{V_{t+1}^{\text{Ret}}(M_{t+1}^j(A_i)), V_{t+1}^{W}(M_{t+1}^j(A_i), \xi_k)\}$

        Compute $\mathbb{1}_{t+1}^0(M_{t+1}^j(A_i), \xi_k) = 1$ if $V_{t+1}^{\text{Ret}}(M_{t+1}^j(A_i)) \geq V_{t+1}^{W}(M_{t+1}^j(A_i), \xi_k)$ and 0 otherwise.

        Compute $\mathbb{1}_{t+1}^1(M_{t+1}^j(A_i), \xi_k) = 1 - \mathbb{1}_{t+1}^0(M_{t+1}^j(A_i), \xi_k)$

        Compute $\text{RHS}_1^{\text{pre}}(M_{t+1}^j(A_i), \xi_k) = \mathbb{1}_{t+1}^0(M_{t+1}^j(A_i), \xi_k) \cdot u'(c_{t+1}^{\text{Ret}}(M_{t+1}^j(A_i))) + \xi_k \cdot \mathbb{1}_{t+1}^1(M_{t+1}^j(A_i), \xi_k) \cdot u'(c_{t+1}^{W}(M_{t+1}^j(A_i)))$

        Compute $\text{RHS}_2^{\text{pre}}(M_{t+1}^j(A_i), \xi_k) = \xi_k \cdot u'(c_{t+1}^{W}(M_{t+1}^j(A_i)))$

  Compute $\mathcal{V}_t^1(A_i) = \sum_{j=1}^J \sum_{k=1}^K \omega_j \mu_k \cdot V_{t+1}^{RW}(M_{t+1}^j(A_i), \xi_k)$

  Compute $\mathcal{V}_t^2(A_i) = \sum_{j=1}^J \sum_{k=1}^K \omega_j \mu_k \cdot V_{t+1}^{W}(M_{t+1}^j(A_i), \xi_k)$

  Compute $\varepsilon_t(A_i) = 1 - \exp\left(-\beta(1-p)(\mathcal{V}_t^1(A_i) - \mathcal{V}_t^2(A_i))/\sigma_s\right)$

  Compute $\theta(\varepsilon_t(A_i)) = \sigma_s(\varepsilon_t(A_i) + (1 - \varepsilon_t(A_i))\log(1 - \varepsilon_t(A_i)))$

  Compute $P_1(\varepsilon_t(A_i)) = (1 - \varepsilon_t(A_i)) \cdot p + \varepsilon_t(A_i)$

  Compute $P_2(\varepsilon_t(A_i)) = (1 - \varepsilon_t(A_i)) \cdot (1 - p)$

  Compute $\text{RHS}_1(A_i) = \sum_{j=1}^J \sum_{k=1}^K \omega_j \mu_k \cdot \text{RHS}_1^{\text{pre}}(M_{t+1}^j(A_i), \xi_k)$

  Compute $\text{RHS}_2(A_i) = \sum_{j=1}^J \sum_{k=1}^K \omega_j \mu_k \cdot \text{RHS}_2^{\text{pre}}(M_{t+1}^j(A_i), \xi_k)$

  Compute $\text{RHS}(A_i) = R\beta(P_1(\varepsilon_t(A_i))\text{RHS}_1(A_i) + P_2(\varepsilon_t(A_i))\text{RHS}_2(A_i))$

  Compute expected value function $\mathbb{E}V_{t+1}(M_{t+1}(A_i)) = P_1(\varepsilon_t(A_i)) \cdot \mathcal{V}_t^1(A_i) + P_2(\varepsilon_t(A_i)) \cdot \mathcal{V}_t^2(A_i)$

  Compute current consumption $c_t^W(A_i) = u'^{-1}(\text{RHS}(A_i))$

  Compute value function $V_t^W(M_t(A_i)) = u(c_t^W(A_i), 1, \varepsilon_t(A_i)) + \beta\mathbb{E}V_{t+1}(M_{t+1}(A_i))$

  Compute endogenous grid point $M_t(A_i) = c_t^W(A_i) + A_i$

Collect the points $M_t(A_i)$, $c_t^W(A_i)$, and $V_t^W(M_t(A_i))$ to form the consumption function, $c_t^W(\vec{M}_t)$, and value function, $V_t^W(\vec{M}_t)$.

Rationale. Because Π is assembled once and reused across periods, the per-period cost of the income expectation reduces to a matrix multiplication. Relative to a coarse equiprobable shock grid, the matrix integrates the continuous density and thereby avoids the phantom kinks documented in Section 2.5 (Figure 7), typically at lower cost than pushing discretization to a very fine N (Table 1). As noted there, a fixed (b, m) grid may need widening if σ_η is scaled up substantially.

3 Durable model

The second example is a consumption-savings model with durable and nondurable goods. It is a simplified version of the model in Monninger [2025] which uses endogeneous search as a numerical tool to solve the model. This section focuses on the solution method and compares the usage of taste shocks, endogeneous search, and stochastic durable prices.

3.1 Model setup

The model consists of ex-ante identical agents who face uninsurable income risk and can consume non-durable goods, receive flow utility from a durable good, and can save in one single risk-free liquit asset.

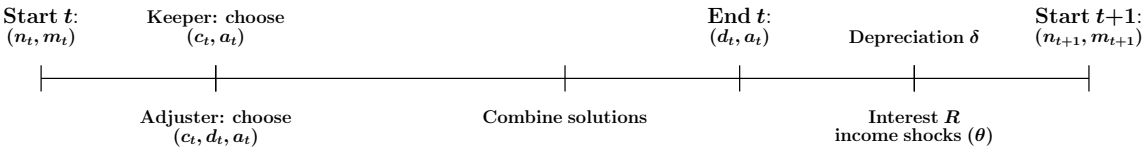


Figure 8: Timeline of the durable model

The agent starts each period with beginning-of-period durable stock (n_t) and market resources (m_t). The agent solves the adjuster and keeper problems separately and chooses optimal levels of nondurable consumption c_t, durable stock d_t, and assets a_t. The discrete keep-versus-adjust decision is then resolved by a hard maximum over the two conditional value functions. At the end of the period, the agent holds a durable stock d_t and liquid assets a_t. Between period t and $t+1$, the transitory income shock materializes, interest is received, and the durable stock depreciates.

Each household is characterized by two state variables: beginning-of-period market resources m_t and the stock of durables n_t. Preferences exhibit constant relative risk aversion and combine nondurable consumption c_t and the service flow of durables d_t using a Cobb-Douglas aggregator[1]:

$$u(c_t, d_t) = \frac{\left(c_t^\alpha d_t^{1-\alpha}\right)^{1-\rho}}{1-\rho}, \qquad \alpha \in (0,1),\ \rho > 0. \tag{28}$$

At the start of the period, the household observes (m_t, n_t) and decides whether to keep the durable stock unchanged or to adjust. If no adjustment occurs, the household sets $d_t = n_t$ and allocates m_t between nondurable consumption and savings. If the household adjusts, it sells the entire beginning-of-period stock at no price risk but incurs a proportional adjustment cost $\tau \in (0,1)$. The resulting liquidity is

$$\ell_t = m_t + (1-\tau)n_t, \tag{29}$$

[1]Using this form is standard (eg see Berger and Vavra [2015], McKay and Wieland [2021]) which is in line with micro-evidence Berger et al. [2018]. An alternative approach is to use a CES aggregator as in Mankiw [1983]

which can then be allocated across nondurable consumption, a new durable stock d_t, and savings a_t.

The household's problem is the hard discrete choice between keeping and adjusting:

$$v_t(m_t, n_t) = \max\left\{v_t^{\text{keep}}(m_t, n_t),\ v_t^{\text{adj}}(\ell_t)\right\}, \tag{30}$$

where v_t^{keep} and v_t^{adj} are the value functions conditional on keeping or adjusting.

Labor income is idiosyncratic and purely transitory. Normalized income is

$$y_t = \theta_t, \tag{31}$$

where θ_t is i.i.d. log-normally distributed with $\mathbb{E}[\theta_t] = 1$ and a strictly positive standard deviation. There is no unemployment risk.

Assets earn a single interest rate r, so the gross interest factor is $\mathcal{R} = 1 + r$. The durable stock depreciates at rate $\delta \in (0, 1)$. There are no maintenance costs and no operation costs. Therefore, next-period states evolve according to

$$m_{t+1} = \mathcal{R}a_t + \theta_{t+1}, \tag{32}$$
$$n_{t+1} = (1 - \delta)\, d_t. \tag{33}$$

Borrowing is prohibited: the loan-to-value ratio is zero and the lower bound on assets is $\underline{a} = 0$, so end-of-period assets must satisfy $a_t \geq 0$.

Hence, the Bellman equations conditional on keep and adjust are:

$$v_t^{\text{keep}}(m_t, n_t) = \max_{c_t}\ u(c_t, d_t) + \beta\mathbb{E}_t\big[v_{t+1}(m_{t+1}, n_{t+1})\big], \tag{34}$$
$$\text{s.t. } a_t = m_t - c_t \geq 0, \tag{35}$$
$$d_t = n_t, \tag{36}$$
$$m_{t+1} = \mathcal{R}a_t + \theta_{t+1}, \tag{37}$$
$$n_{t+1} = (1 - \delta)\, d_t. \tag{38}$$

$$v_t^{\text{adj}}(\ell_t) = \max_{c_t, d_t}\ u(c_t, d_t) + \beta\mathbb{E}_t\big[v_{t+1}(m_{t+1}, n_{t+1})\big], \tag{39}$$
$$\text{s.t. } a_t = \ell_t - c_t - d_t \geq 0, \tag{40}$$
$$m_{t+1} = \mathcal{R}a_t + \theta_{t+1}, \tag{41}$$
$$n_{t+1} = (1 - \delta)\, d_t. \tag{42}$$

With the following first-order conditions with respect to nondurable consumption and durable stock:

$$\frac{\partial v_t^{adj}}{\partial c_t} = u_c(c_t, d_t) \geq \beta\mathcal{R}\,\mathbb{E}_t\big[v_{t+1}^m\big], \tag{43}$$
$$\frac{\partial v_t^{adj}}{\partial d_t} = u_d(c_t, d_t) \geq \beta\mathcal{R}\,\mathbb{E}_t\big[v_{t+1}^m\big] - \beta(1 - \delta)\,\mathbb{E}_t\big[v_{t+1}^n\big], \tag{44}$$

Equations 43 and 44 are stated as KKT conditions. Equality holds when the borrowing constraint does not bind, and the left-hand side strictly exceeds the right-hand side when it does. Equation 43 equates the marginal utility of nondurable consumption with the discounted shadow value of savings. Equation 44 shows that the marginal utility of durables equals the net user cost: the first term on the right is the discounted opportunity cost $\mathcal{R}$ of diverting a unit of resources into durables rather than liquid assets, and the second term is the discounted benefit of carrying the remaining durable stock $(1 - \delta)d_t$ into the next period.

3.2 Algorithm

This section outlines the algorithm for one recursive period of the baseline model: no Extreme Value Type I taste shocks, no directed search, and no durable-price dispersion. The discrete keep-versus-adjust choice is resolved by a hard maximum. The terminal period is closed form—the keeper sets $c_T = m_T$ and $d_T = n_T$, the adjuster sets $c_T = \alpha \ell_T$ and $d_T = (1-\alpha)\ell_T$, and the two branches are combined by a hard max—so Algorithm 2 starts from a given next-period solution $(v_{t+1},\, v^m_{t+1},\, v^n_{t+1})$.

Define the end-of-period (gothic) objects on (d_t, a_t) by

$$\mathfrak{v}'_c(d_t, a_t) = \beta\, \mathcal{R}\, \mathbb{E}_t\big[v^m_{t+1}(n_{t+1}, m_{t+1})\big], \tag{45}$$

$$\mathfrak{v}'_d(d_t, a_t) = \beta\, (1-\delta)\, \mathbb{E}_t\big[v^n_{t+1}(n_{t+1}, m_{t+1})\big], \tag{46}$$

$$\mathfrak{v}(d_t, a_t) = \beta\, \mathbb{E}_t\big[v_{t+1}(n_{t+1}, m_{t+1})\big], \tag{47}$$

with $m_{t+1} = \mathcal{R} a_t + \theta_{t+1}$ and $n_{t+1} = (1-\delta) d_t$. The expectation over θ_{t+1} may be evaluated by quadrature (as written below) or by the transition-matrix integration described in Appendix A. Because adjustment costs make the first-order conditions only necessary, the keeper step applies a DC-EGM upper envelope [Iskhakov et al., 2017] and the adjuster step applies FUES [Dobrescu and Shanker, 2022].

3.3 Illustration of the numerical challenge

Fig. 9 illustrates the terminal solution. Taking the hard-max between the options of adjusting and keeping, the S,s structure emerges (fig. 9a) which leads to a sharp switch between fully adjusting or fully keeping. As a result, the optimal policy functions of the model conditional on adjusting and keeping experience discontinuities along the switchpoints as illustrated for the durable consumption function (fig. 9b).

In earlier period, the agent incorporates the switchpoints in the next period into the solution of today's problem. Together with non-convex adjustment costs, the first order conditions are only necessary, but not sufficient. Hence, we need to incorporate an upper envelope algorithm such as FUES. However, even with this algorithm, small discontinuities still arise in the adjuster policy functions. Additionally, by combining the keeper and adjuster functions each period, these non-convexities propagade backwards similar to the retirement problem. Figure 10 illustrates this for a fixed beginning-of-period durable stock n_t. In the yellow adjuster region, the policy function and MPC follows the adjuster policy function. With higher market resources, the S,s border gets crossed resulting in stark spikes in MPCs.

3.4 Extension: Extreme Value Type I Taste Shocks

The hard maximum in the discrete keep-versus-adjust decision produces a kink in the value function and a discontinuity in the policy functions at the state combinations (m_t, n_t) where the household is indifferent between the two branches. Following the discrete–continuous choice literature (e.g. Iskhakov et al. [2017]), this subsection smooths that discontinuity with i.i.d. Extreme Value Type I (Gumbel) taste shocks on the two discrete alternatives, while leaving search frictions aside (these are introduced separately in Section 3.5). Unlike the multiplicative mean-one log-normal taste shocks used in the retirement application of Section 2, EV Type I shocks enter *additively* on each branch and are integrated out in closed form: they randomize the discrete outcome directly through logit choice probabilities, without introducing an additional continuous (or discretized) state.

Algorithm 2: EGM step for the baseline durable model (hard max)

Let $\vec{\theta} = \{\theta_1, \ldots, \theta_J\}$ be quadrature points for the transitory income shock with weights $\vec{\omega} = \{\omega_1, \ldots, \omega_J\}$.

Form ascending grids $\vec{a}_t = \{a^1, \ldots, a^I\}$, $\vec{d}_t = \{d^1, \ldots, d^L\}$, a fine durable grid $\vec{d}_t^{\,\text{adj}}$ for the adjuster FOC scan, and beginning-of-period grids $\vec{n}_t$, $\vec{m}_t$.

`/* End-of-period (gothic) functions */`

for $i = 1, \ldots, I$ **do**

  for $q = 1, \ldots, L$ **do**

    for $j = 1, \ldots, J$ **do**

      Compute $m_{t+1}^j(a^i) = \mathcal{R}a^i + \theta_j$ and $n_{t+1}(d^q) = (1-\delta)d^q$

      Interpolate v_{t+1}, v_{t+1}^m, and v_{t+1}^n at $(n_{t+1}(d^q),\, m_{t+1}^j(a^i))$

    Compute $\mathfrak{v}'_c(d^q, a^i) = \beta\mathcal{R}\sum_{j=1}^J \omega_j\, v_{t+1}^m(n_{t+1}(d^q), m_{t+1}^j(a^i))$

    Compute $\mathfrak{v}'_d(d^q, a^i) = \beta(1-\delta)\sum_{j=1}^J \omega_j\, v_{t+1}^n(n_{t+1}(d^q), m_{t+1}^j(a^i))$

    Compute $\mathfrak{v}(d^q, a^i) = \beta\sum_{j=1}^J \omega_j\, v_{t+1}(n_{t+1}(d^q), m_{t+1}^j(a^i))$

`/* Keeper: EGM with` $d_t = n_t$`, then DC-EGM upper envelope */`

for *each* $n \in \vec{n}_t$ **do**

  for $i = 1, \ldots, I$ **do**

    Compute $c_t^{\text{keep}}(a^i; n) = u_c^{-1}\big(\mathfrak{v}'_c(n, a^i);\, n\big)$

    Compute endogenous resources $m_t^{\text{EGM}}(a^i; n) = a^i + c_t^{\text{keep}}(a^i; n)$

    Compute $v_t^{\text{EGM}}(a^i; n) = u\big(c_t^{\text{keep}}(a^i; n), n\big) + \mathfrak{v}(n, a^i)$

  Apply a DC-EGM upper envelope to $\{(m_t^{\text{EGM}},\, c_t^{\text{keep}},\, v_t^{\text{EGM}})\}$ and interpolate onto $\vec{m}_t$ to obtain $c_t^{\text{keep}}(n, m)$, $a_t^{\text{keep}} = \max\{m - c_t^{\text{keep}}, 0\}$, and $d_t^{\text{keep}} = n$

  Compute $v_t^{\text{keep}}(n, m) = u(c_t^{\text{keep}}, n) + \mathfrak{v}(n, a_t^{\text{keep}})$, $v_t^{m,\text{keep}} = u_c(c_t^{\text{keep}}, n)$, and $v_t^{n,\text{keep}} = u_d(c_t^{\text{keep}}, n) + \mathfrak{v}'_d(n, a_t^{\text{keep}})$

`/* Adjuster: FOC roots in` a `on a fine` d`-grid, then FUES in liquidity */`

Initialize empty candidate clouds for $(\ell_t, c_t, d_t, a_t, v_t)$

for *each* $d \in \vec{d}_t^{\,\text{adj}}$ **do**

  Find all roots a of $u_d(c, d) - \mathfrak{v}'_c(d, a) + \mathfrak{v}'_d(d, a) = 0$ with $c = u_c^{-1}\big(\mathfrak{v}'_c(d, a);\, d\big)$

  for *each root* a **do**

    Compute $\ell_t = a + c + d$ and $v_t^{\text{adj}} = u(c, d) + \mathfrak{v}(d, a)$

    Append $(\ell_t, c, d, a, v_t^{\text{adj}})$ to the candidate cloud

Sort candidates by ℓ_t and apply FUES to obtain refined points $(\ell_t, c_t^{\text{adj}}, d_t^{\text{adj}}, a_t^{\text{adj}}, v_t^{\text{adj}})$

Construct one-dimensional interpolants $c_t^{\text{adj}}(\ell)$, $d_t^{\text{adj}}(\ell)$, and $a_t^{\text{adj}}(\ell)$

for *each* $(n, m) \in \vec{n}_t \times \vec{m}_t$ **do**

  Compute $\ell_t = m + (1-\tau)n$

  Evaluate $c_t^{\text{adj}}(\ell_t)$, $d_t^{\text{adj}}(\ell_t)$, and $a_t^{\text{adj}} = \max\{\ell_t - c_t^{\text{adj}} - d_t^{\text{adj}}, 0\}$

  Compute $v_t^{\text{adj}}(\ell_t) = u(c_t^{\text{adj}}, d_t^{\text{adj}}) + \mathfrak{v}(d_t^{\text{adj}}, a_t^{\text{adj}})$, $v_t^{m,\text{adj}} = u_c(c_t^{\text{adj}}, d_t^{\text{adj}})$, and $v_t^{n,\text{adj}} = (1-\tau)\, u_c(c_t^{\text{adj}}, d_t^{\text{adj}})$

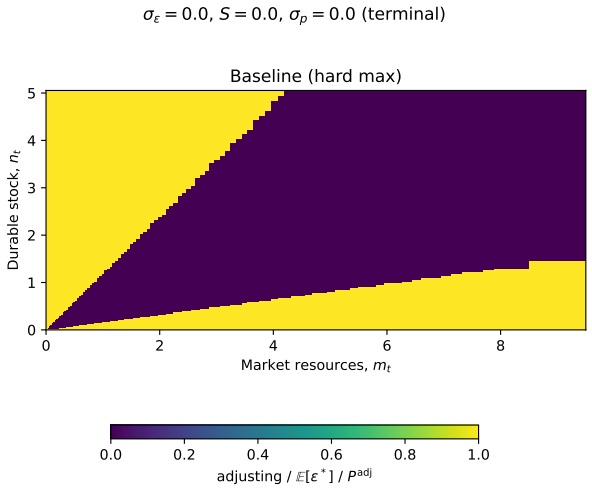
`/* Hard-max discrete choice */`

for *each* $(n, m) \in \vec{n}_t \times \vec{m}_t$ **do**

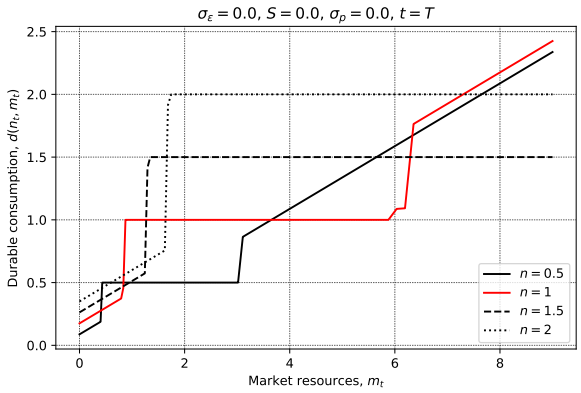
  Set $\mathcal{I}_t(n, m) = 1$ if $v_t^{\text{adj}}\big(m + (1-\tau)n\big) \geq v_t^{\text{keep}}(n, m)$ and 0 otherwise

  Set $(c_t, d_t, a_t, v_t, v_t^m, v_t^n)$ to the adjuster branch if $\mathcal{I}_t = 1$ and to the keeper branch if $\mathcal{I}_t = 0$

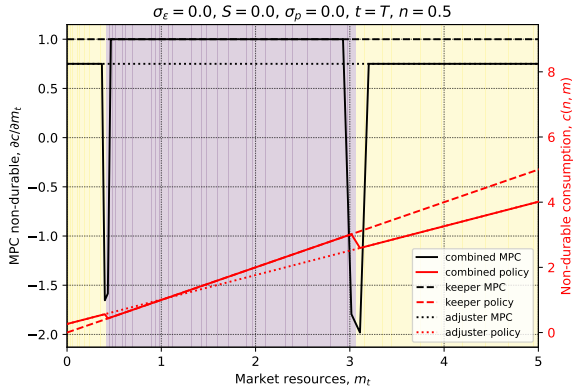
Collect the combined policies and (inverse) value functions on $(\vec{n}_t, \vec{m}_t)$ for use in period $t-1$.



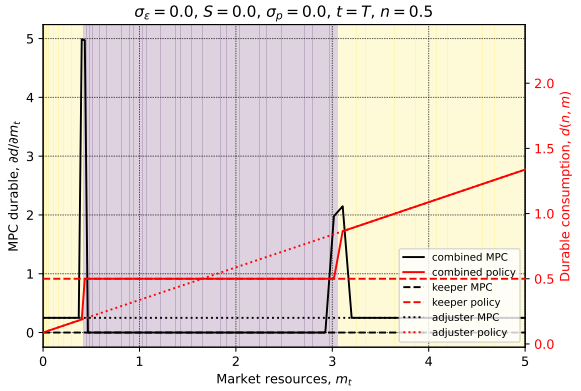
(a) Adjustment region (S,s bands).



(b) Durable policy $d_t(n_t, m_t)$.

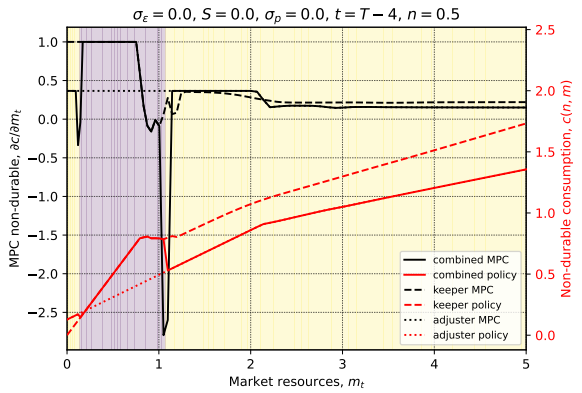


(c) MPC decomposition of nondurable consumption $c_t(n_t, m_t)$.

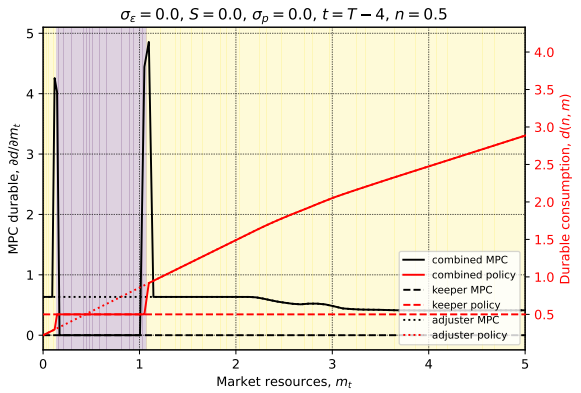


(d) MPC decomposition of durable consumption $d_t(n_t, m_t)$.

Figure 9: Terminal-period solution of the baseline durable model (hard max). The upper left panel shows the keep–adjust regions in (m_t, n_t) space; the upper right panel shows the combined durable policy for selected beginning-of-period durable stocks n_t. The lower panels show the MPC decomposition of nondurable and durable consumption given a fixed beginning-of-period durable stock n_t.



(a) MPC decomposition of nondurable consumption $c_t(n_t, m_t)$.



(b) MPC decomposition of durable consumption $d_t(n_t, m_t)$.

Figure 10: MPC and policy functions of non-durable and durable consumption.

At the start of period t, after observing (m_t, n_t), the household draws i.i.d. Extreme Value Type I shocks $(\varepsilon_t^{\text{keep}}, \varepsilon_t^{\text{adj}})$ and chooses the branch that maximizes the shock-augmented conditional value,

$$v_t(m_t, n_t) = \max\left\{v_t^{\text{keep}}(m_t, n_t) + \sigma_\varepsilon\, \varepsilon_t^{\text{keep}},\ v_t^{\text{adj}}(\ell_t) + \sigma_\varepsilon\, \varepsilon_t^{\text{adj}}\right\}, \tag{48}$$

where $\sigma_\varepsilon > 0$ is the taste-shock scale and $\ell_t = m_t + (1-\tau)n_t$. The conditional value functions $v_t^{\text{keep}}(m_t, n_t)$ and $v_t^{\text{adj}}(\ell_t)$ retain exactly the same Bellman definitions as in the baseline model: the taste shocks do not enter the inner continuous maximizations, so the first-order conditions (43)–(44) and the continuous policies c_t^{keep}, c_t^{adj}, d_t^{adj} are unchanged.

Because $(\varepsilon_t^{\text{keep}}, \varepsilon_t^{\text{adj}})$ are i.i.d. Extreme Value Type I, the expectation of (48) has the closed-form log-sum representation

$$v_t(m_t, n_t) = \sigma_\varepsilon \log\left(\exp\left(v_t^{\text{keep}}(m_t, n_t)/\sigma_\varepsilon\right) + \exp\left(v_t^{\text{adj}}(\ell_t)/\sigma_\varepsilon\right)\right), \tag{49}$$

with logit choice probabilities

$$P_t^{\text{adj}}(m_t, n_t) = \frac{\exp\left(v_t^{\text{adj}}(\ell_t)/\sigma_\varepsilon\right)}{\exp\left(v_t^{\text{keep}}(m_t, n_t)/\sigma_\varepsilon\right) + \exp\left(v_t^{\text{adj}}(\ell_t)/\sigma_\varepsilon\right)}, \qquad P_t^{\text{keep}}(m_t, n_t) = 1 - P_t^{\text{adj}}(m_t, n_t). \tag{50}$$

The expected (ex-ante) policy functions are the probability-weighted averages of the two branches,

$$c_t(m_t, n_t) = P_t^{\text{keep}}\, c_t^{\text{keep}}(m_t, n_t) + P_t^{\text{adj}}\, c_t^{\text{adj}}(\ell_t), \tag{51}$$

$$d_t(m_t, n_t) = P_t^{\text{keep}}\, n_t + P_t^{\text{adj}}\, d_t^{\text{adj}}(\ell_t). \tag{52}$$

As $\sigma_\varepsilon \to 0$, the logit probabilities collapse to the hard indicators of the baseline model and (49) collapses to the hard maximum. Larger σ_ε spreads probability mass onto the inferior branch near indifference, so c_t and d_t (and the marginal-value functions derived from the inclusive value (49)) are smooth in (m_t, n_t). No quadrature over taste nodes is required: the EV Type I assumption delivers the inclusive value and choice probabilities in closed form. The taste shocks do not alter the household's budget set, so the no-borrowing constraint $a_t \geq 0$ is unaffected. I derive the

envelope conditions, the Euler equations, and the EGM algorithm with EV Type I taste shocks in appendix A.2.

As a result, the S,s curve in the terminal period is smoothened and adjuster values between zero and one are optimal. In the aggregate, the policy functions are a convex combination of adjuster and keeper policy functions around the S,s thresholds. Fig 11 illustrates the point clearly.

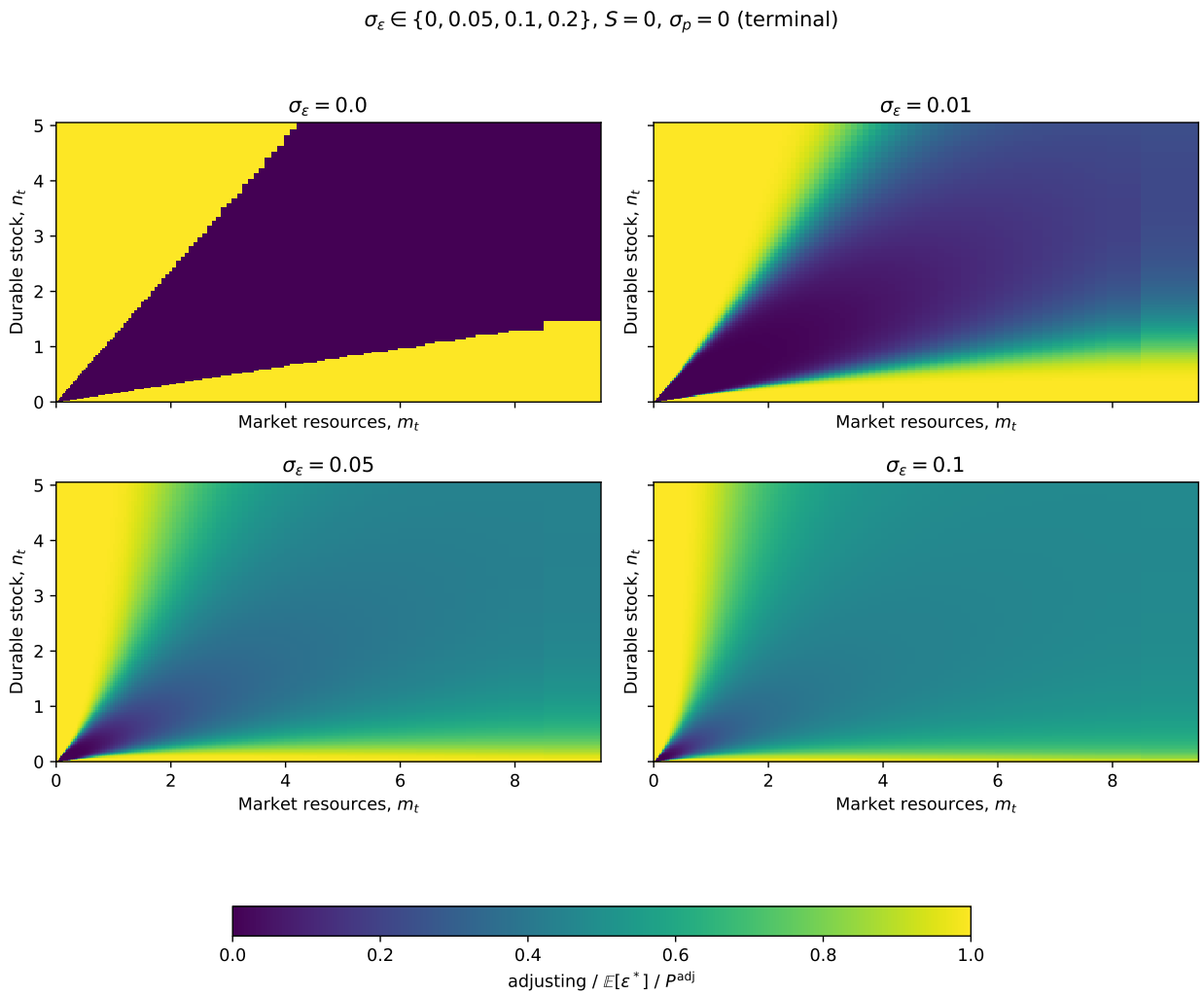
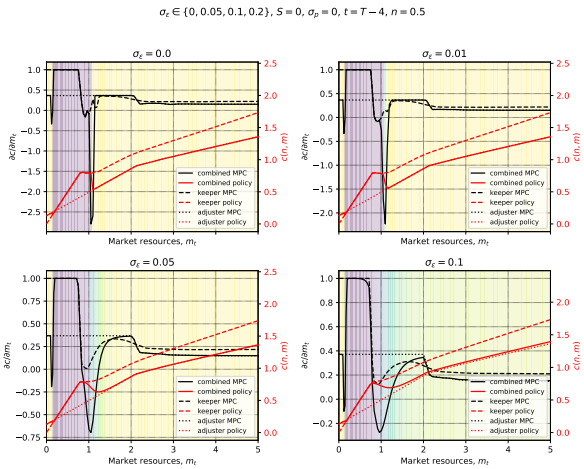


Figure 11: Terminal-period keep–adjust regions in (m_t, n_t) space under EV Type I taste shocks for $\sigma_\varepsilon \in \{0, 0.05, 0.1, 0.2\}$.

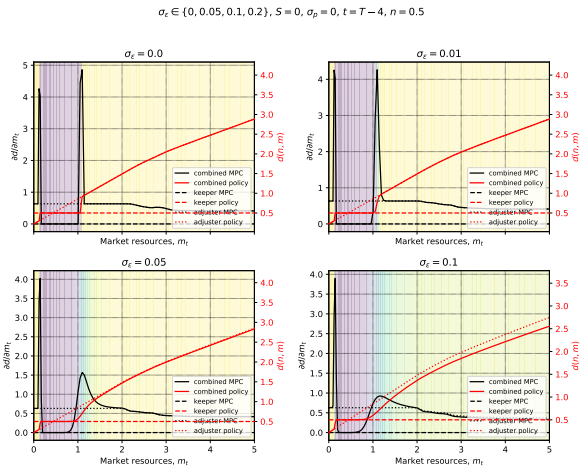
As a result, the discontinuities in period t disapear with a small EV shock parameter. Fig 12 show the now continuous consumption functions and marginal propensities to consume and expend.

3.5 Extension: Search Frictions

This subsection instead reintroduces the directed-search mechanism of the full model into the basic environment, without taste shocks. Rather than resolving the keep-versus-adjust margin with a hard maximum, the household chooses a search intensity $\varepsilon_t \in [0, 1]$ that determines the weight placed on the adjuster's conditional value in the current period. Exerting search effort is costly,



(a) MPC decomposition of nondurable consumption $c_t(n_t, m_t)$.



(b) MPC decomposition of durable consumption $d_t(n_t, m_t)$.

Figure 12: MPC and policy functions of non-durable and durable consumption using EV taste shocks.

with cost function

$$\sigma(\varepsilon_t) = S\big[\varepsilon_t + (1-\varepsilon_t)\log(1-\varepsilon_t)\big], \tag{53}$$

where $S > 0$ governs the degree of friction in the market for durables, following Khan [2019]. The beginning-of-period problem becomes

$$v_t(m_t, n_t) = \max_{\varepsilon_t \in [0,1]} \Big[(1-\varepsilon_t)\, v_t^{\text{keep}}(m_t, n_t) + \varepsilon_t\, v_t^{\text{adj}}(\ell_t) - \sigma(\varepsilon_t)\Big], \tag{54}$$

where v_t^{keep} and v_t^{adj} retain exactly the same Bellman definitions as in the baseline model, except that the continuation value $v_{t+1}(m_{t+1}, n_{t+1})$ appearing inside them is now itself given recursively by equation (54) one period ahead, rather than by the hard maximum of the baseline model. All other features of the basic environment—the single interest rate $\mathcal{R}$, the absence of maintenance costs, and the no-borrowing constraint $a_t \geq 0$—are unchanged.

Search effort is chosen before the discrete outcome is known: ε_t can be interpreted as the probability with which the household successfully locates a suitable durable this period, so the combined value function (54) is a probability-weighted average of the two conditional values, net of the search cost. As $S \to 0$, search becomes costless, the household sets $\varepsilon_t \in \{0, 1\}$, and the model collapses to the hard-maximum baseline; as $S \to \infty$, search becomes prohibitively costly, $\varepsilon_t^* \to 0$, and the household never adjusts. I derive the optimal search-effort rule, the smooth combined marginal-value functions, and the associated EGM steps in appendix A.3.

Endogeneous search achieves a similar smoothing compared to the EV shocks. Fig 13 illustrates the terminal period solution. A small search cost is sufficient to lead to a smoothened S,s region.

As a result, optimal consumption functions in period t are smooth as well (see fig 14).

3.6 Extension: Stochastic Durable Prices

This subsection introduces a third source of variation, orthogonal to the taste shocks of Section 3.4 and the search frictions of Section 3.5: an idiosyncratic, i.i.d. shock to the relative price at which

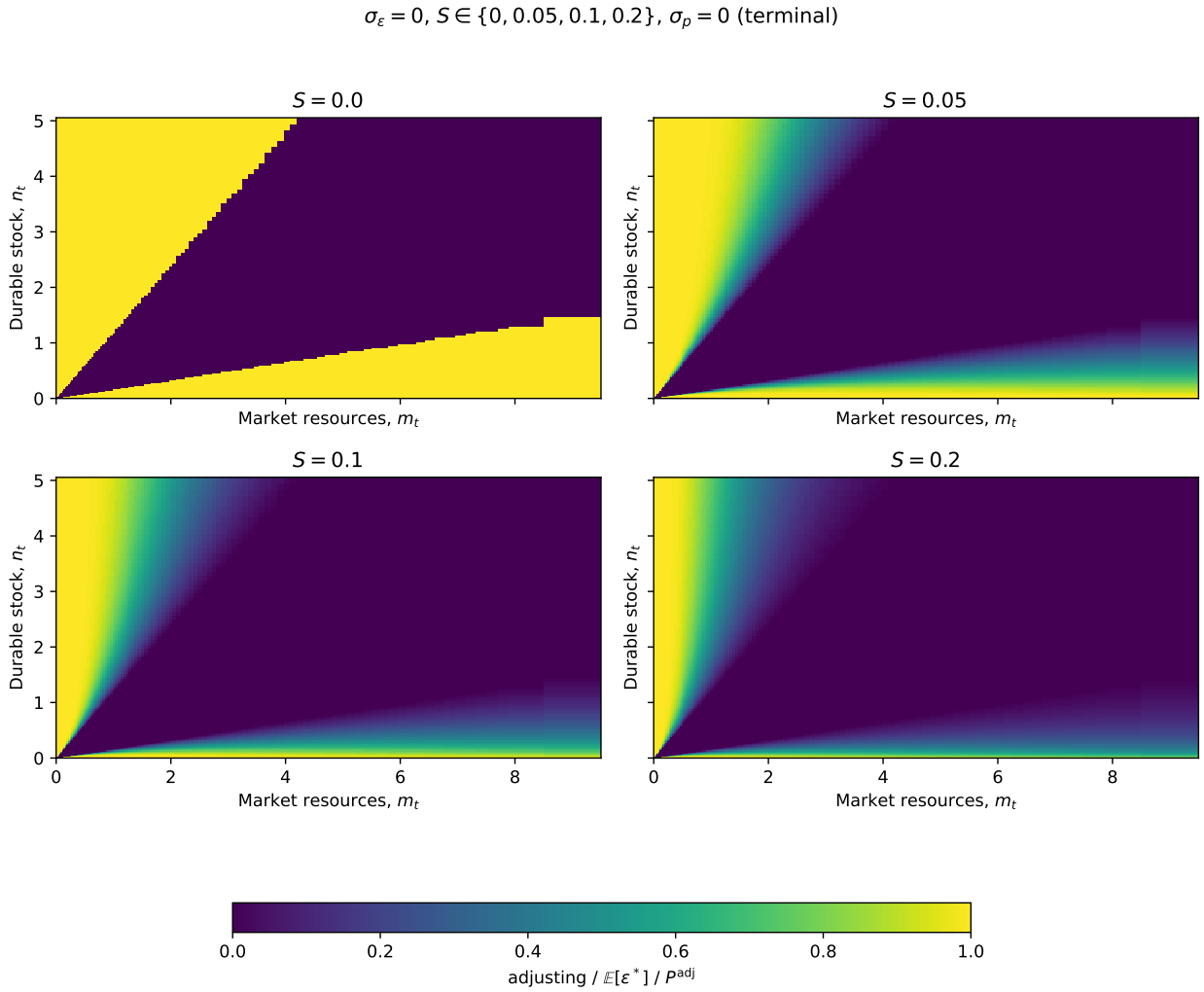
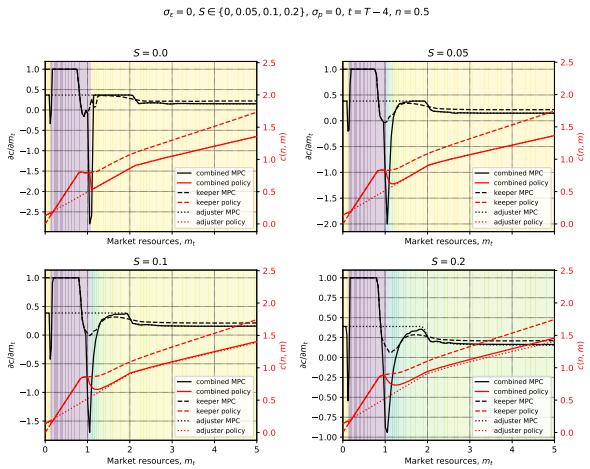
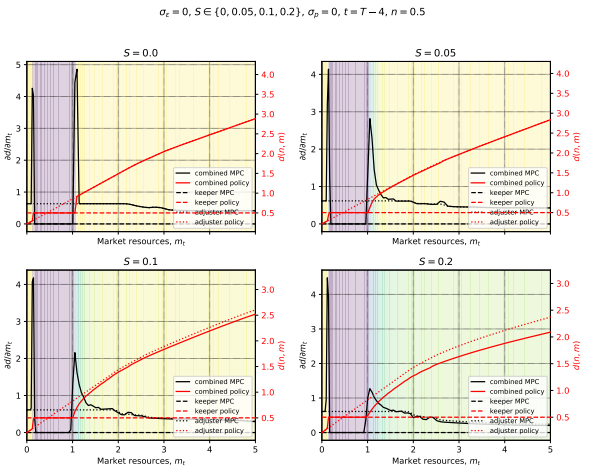


Figure 13: Terminal-period keep–adjust regions in (m_t, n_t) space under endogeneous search for search-cost scales $S \in \{0, 0.05, 0.1, 0.2\}$.



(a) MPC decomposition of nondurable consumption $c_t(n_t, m_t)$.



(b) MPC decomposition of durable consumption $d_t(n_t, m_t)$.

Figure 14: MPC and policy functions of non-durable and durable consumption using endogeneous search.

the durable good can be purchased in the period of adjustment. Neither taste shocks nor search frictions are present in this subsection, so the discrete choice between keeping and adjusting again resolves through a hard maximum, but now the adjuster's problem—and hence the comparison itself—is contingent on the realized price.

At the start of period t, after the transitory income shock θ_t has materialized, the household observes an i.i.d. relative price p_t at which new units of the durable good can be bought this period. The price is log-normally distributed with mean one,

$$\log p_t \sim \mathcal{N}\Big(-\frac{\sigma_p^2}{2}, \sigma_p^2 \Big), \qquad \mathbb{E}[p_t] = 1, \tag{55}$$

independent of θ_t and independent across periods. Only the *purchase* of new durables is exposed to this price risk: households liquidate their existing durable stock exactly as in the baseline model, at the fixed proportional recovery rate $1 - \tau$, so the liquidity available upon adjustment, $\ell_t = m_t + (1 - \tau)n_t$, is unaffected by p_t. This is consistent with an interpretation in which trading in a used durable is governed by a separate, deterministic adjustment cost, while the price of a new unit on the market this period fluctuates, e.g. a household trading in a used car against the fluctuating price of a new one.

The keeper's problem is entirely unaffected by p_t, since no purchase occurs, so $v_t^{\text{keep}}(m_t, n_t)$ retains its baseline Bellman definition. The adjuster's budget constraint, however, now reads

$$a_t = \ell_t - c_t - p_t\, d_t \geq 0, \tag{56}$$

in place of $a_t = \ell_t - c_t - d_t$, so the conditional value function becomes a function of both ℓ_t and p_t,

$$v_t^{\text{adj}}(\ell_t, p_t) = \max_{c_t, d_t} \; u(c_t, d_t) + \beta \mathbb{E}_t\big[v_{t+1}(m_{t+1}, n_{t+1})\big], \tag{57}$$

$$\text{s.t. equation (56)}, \tag{58}$$

$$m_{t+1} = \mathcal{R} a_t + \theta_{t+1}, \qquad n_{t+1} = (1 - \delta)\, d_t. \tag{59}$$

Everything else about the household's environment—the interest rate $\mathcal{R}$, the depreciation rate δ, and the no-borrowing constraint $a_t \geq 0$—is unchanged, and next period's durable stock n_{t+1} continues to be measured in physical units, so p_t affects only the current-period cost of acquiring d_t, not its future service flow beyond ordinary depreciation.

With the price-contingent budget constraint (56), the adjuster's Lagrangian is

$$\mathscr{L} = u(c_t, d_t) + \beta \mathbb{E}_t\big[v_{t+1}(m_{t+1}, n_{t+1})\big] + \lambda_t^{\text{adj}}\big(\ell_t - c_t - p_t\, d_t - a_t\big), \tag{60}$$

with $m_{t+1} = \mathcal{R}a_t + \theta_{t+1}$ and $n_{t+1} = (1-\delta)d_t$. The first-order conditions with respect to c_t, a_t, and d_t are

$$\frac{\partial \mathscr{L}}{\partial c_t} = u_c(c_t, d_t) - \lambda_t^{\text{adj}} = 0 \qquad \Longrightarrow \qquad \lambda_t^{\text{adj}} = u_c(c_t, d_t), \tag{61}$$

$$\frac{\partial \mathscr{L}}{\partial a_t} = \beta\mathcal{R}\,\mathbb{E}_t\big[v_{t+1}^m\big] - \lambda_t^{\text{adj}} = 0 \qquad \Longrightarrow \qquad \lambda_t^{\text{adj}} = \beta\mathcal{R}\,\mathbb{E}_t\big[v_{t+1}^m\big], \tag{62}$$

$$\frac{\partial \mathscr{L}}{\partial d_t} = u_d(c_t, d_t) + \beta(1-\delta)\,\mathbb{E}_t\big[v_{t+1}^n\big] - p_t\,\lambda_t^{\text{adj}} = 0 \qquad \Longrightarrow \qquad u_d(c_t, d_t) = p_t\,\lambda_t^{\text{adj}} - \beta(1-\delta)\,\mathbb{E}_t\big[v_{t+1}^n\big]. \tag{63}$$

In KKT form, these are the price-modified analogues of the baseline conditions (43)–(44):

$$u_c(c_t, d_t) \geq \beta\mathcal{R}\,\mathbb{E}_t\big[v_{t+1}^m\big], \tag{64}$$

$$u_d(c_t, d_t) \geq p_t\,\beta\mathcal{R}\,\mathbb{E}_t\big[v_{t+1}^m\big] - \beta(1-\delta)\,\mathbb{E}_t\big[v_{t+1}^n\big]. \tag{65}$$

Equation (61) is identical in form to (43): because p_t multiplies only d_t in the budget, it does not appear in the c_t first-order condition, so $c_t^{\text{adj},*}$ solves the same equation as in the baseline model *given* d_t. Equation (63) differs from (44): the marginal cost of an extra unit of durables is now $p_t\lambda_t^{\text{adj}}$ rather than λ_t^{adj}, since each unit of d_t costs p_t units of ℓ_t instead of one. Equality holds when the borrowing constraint does not bind.

Because p_t is observed before the discrete choice is made, the household's beginning-of-period problem is the price-contingent hard maximum

$$v_t(m_t, n_t, p_t) = \max\Big\{v_t^{\text{keep}}(m_t, n_t),\ v_t^{\text{adj}}(\ell_t, p_t)\Big\}. \tag{66}$$

A high realization of p_t raises the marginal cost of adjusting and, other things equal, makes the household less likely to adjust and, conditional on adjusting, to purchase a smaller quantity of the durable; a low realization has the opposite effect. Because agents can reject an offer when the drawn price leads to an allocation which is worse than the keeper solution, the model induces state-contingent variation in both the adjustment margin and the size of the resulting durable purchase.

As with the EV Type I inclusive value, the object that matters for the previous period's continuation value is the price-integrated value function

$$\bar{v}_t(m_t, n_t) \equiv \mathbb{E}_{p_t}\Big[v_t(m_t, n_t, p_t)\Big] = \mathbb{E}_{p_t}\Big[\max\big\{v_t^{\text{keep}}(m_t, n_t),\ v_t^{\text{adj}}(\ell_t, p_t)\big\}\Big], \tag{67}$$

which replaces $v_t(m_t, n_t)$ inside the Bellman equations for v_{t-1}^{keep} and v_{t-1}^{adj}. Unlike EV Type I taste shocks, which leave the adjuster's optimal continuous choices unchanged and only reweight the discrete comparison through logit probabilities, the price shock changes the adjuster's optimal consumption-durable split itself, so $v_t^{\text{adj}}(\ell_t, p_t)$ must be solved separately for every relevant realization of p_t; the resulting policy functions $c_t^{\text{adj}}(\ell_t, p_t)$ and $d_t^{\text{adj}}(\ell_t, p_t)$ are likewise price-contingent,

and the model's reported (ex-ante) nondurable-consumption and durable-purchase functions are price-weighted averages of these state-contingent choices, in the same spirit as $\bar{v}_t$. As $\sigma_p \to 0$, $p_t \to 1$ almost surely and the model collapses to the hard-maximum baseline of Section 3.1. I derive the envelope conditions, the terminal-period solution, and the EGM algorithm with stochastic durable prices in appendix A.4.

Even stochastic prices induce enough uncertainty to lead to adjusting intensities between zero and one as seen in fig 15. However, since sharp thresholds still exist, the consumption functions in T-1 are not as smooth as with taste shocks or endogeneous search.

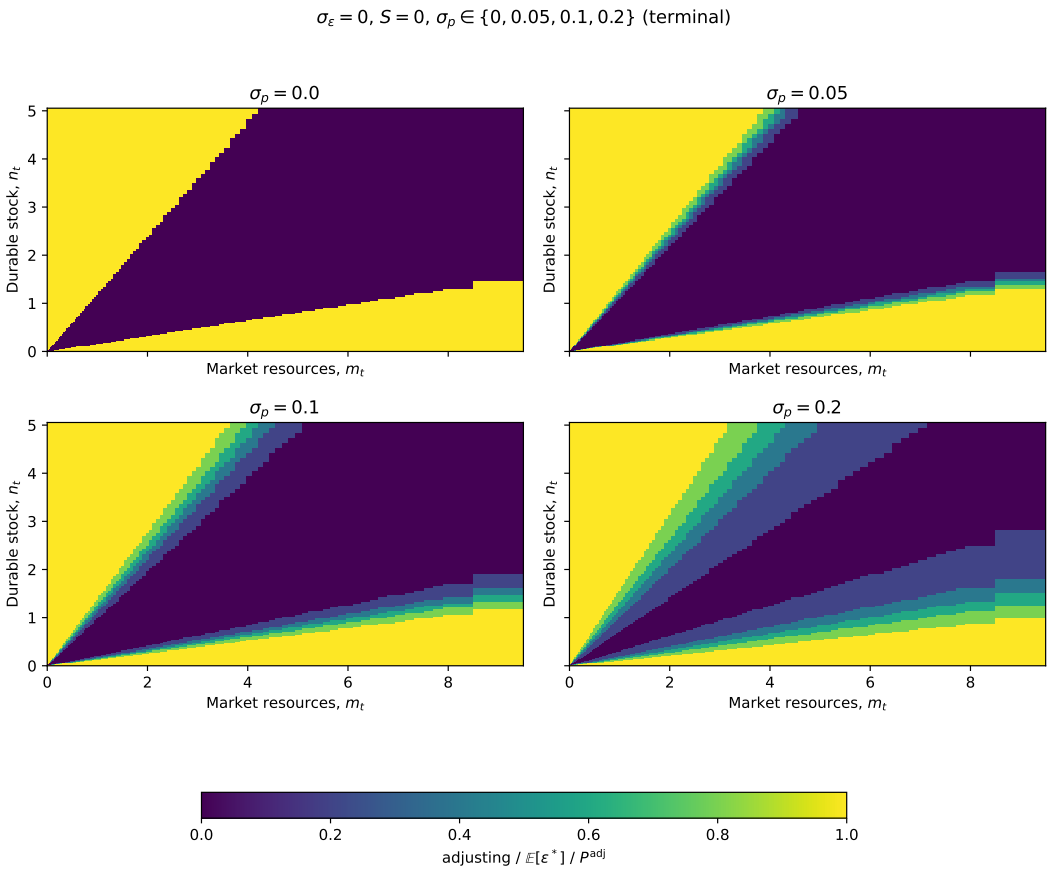


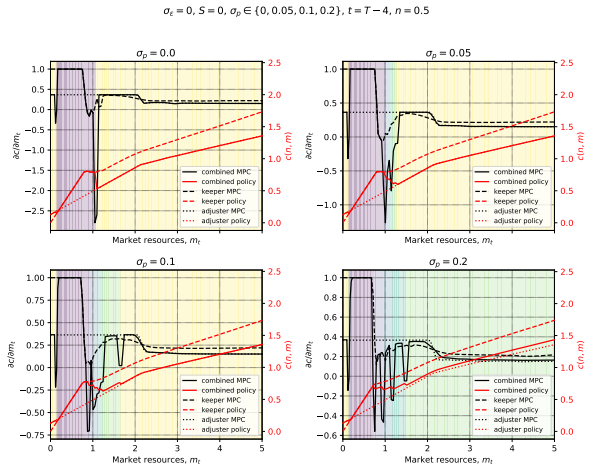
Figure 15: Adjustment regions in period T under stochastic durable prices for $\sigma_p \in \{0, 0.05, 0.1, 0.2\}$ (fixed number of price nodes). Greater price dispersion softens the hard keep–adjust threshold.

Figure 16 shows the policy functions for variation in the price dispersion and figure 17 for a coraser grid of price points.

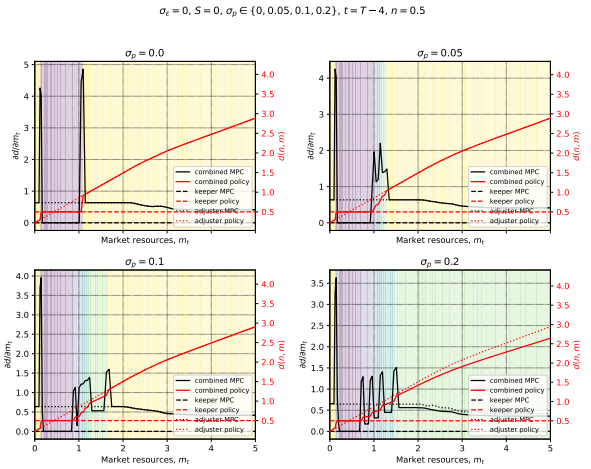
3.7 Extension: Search and Stochastic Prices

This subsection nests the search frictions of Section 3.5 with the stochastic durable prices of Section 3.6. Extreme Value Type I taste shocks (Section 3.4) and directed search are alternative ways of smoothing the keep-versus-adjust margin; nesting both would double-smooth the discrete choice without a clear economic interpretation, so the nested environment uses search rather than EV Type I shocks, together with price risk on the adjuster's continuous problem.

The household faces an idiosyncratic relative price p_t at which new durables can be purchased

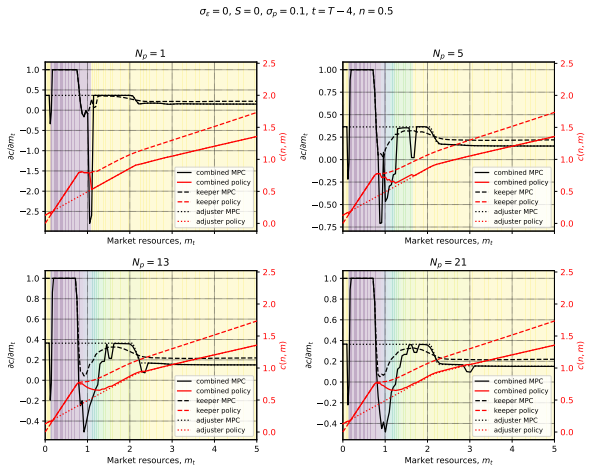


(a) MPC decomposition of nondurable consumption $c_t(n_t, m_t)$.

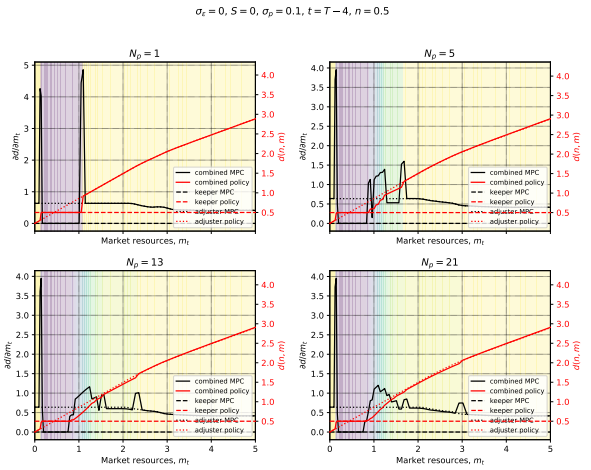


(b) MPC decomposition of durable consumption $d_t(n_t, m_t)$.

Figure 16: MPC and policy functions of non-durable and durable consumption using stochastic prices.



(a) MPC decomposition of nondurable consumption $c_t(n_t, m_t)$.



(b) MPC decomposition of durable consumption $d_t(n_t, m_t)$.

Figure 17: MPC and policy functions of non-durable and durable consumption varying the number of price points.

and chooses a search intensity $\varepsilon_t \in [0,1]$ that determines the probability of successfully adjusting, at convex cost $\sigma(\varepsilon_t)$ from equation (53). The price is independent of income,

$$\log p_t \sim \mathcal{N}\Big(-\frac{\sigma_p^2}{2}, \sigma_p^2\Big), \qquad \mathbb{E}[p_t] = 1, \tag{68}$$

with $p_t \perp \theta_t$ and i.i.d. across periods.

The within-period timeline is as follows. After θ_t has materialized and (m_t, n_t) are known, the household observes p_t. Liquidity upon adjustment remains $\ell_t = m_t + (1-\tau)n_t$, independent of the price. The keeper's problem is unaffected by p_t, so $v_t^{\text{keep}}(m_t, n_t)$ retains its baseline Bellman definition. The adjuster's budget constraint is the price-contingent constraint (56), so the adjuster's conditional value $v_t^{\text{adj}}(\ell_t, p_t)$ and its continuous policies $c_t^{\text{adj}}(\ell_t, p_t)$, $d_t^{\text{adj}}(\ell_t, p_t)$ are exactly as in Section 3.6. Given these two conditional values, the household chooses search intensity to maximize the price-contingent objective

$$v_t(m_t, n_t, p_t) = \max_{\varepsilon_t \in [0,1]} \Big[(1-\varepsilon_t)\, v_t^{\text{keep}}(m_t, n_t) + \varepsilon_t\, v_t^{\text{adj}}(\ell_t, p_t) - \sigma(\varepsilon_t)\Big]. \tag{69}$$

Equation (69) nests the standalone search and price extensions as special cases: with $\sigma_p \to 0$ it collapses to the search problem (54); with $S \to 0$ it collapses to the price-contingent hard maximum (66); and with both limits simultaneous it collapses to the hard-maximum baseline of Section 3.1.

Because p_{t+1} is drawn afresh every period, the object that enters today's Euler equations is the price-integrated value function

$$\bar{v}_t(m_t, n_t) \equiv \mathbb{E}_{p_t}\big[v_t(m_t, n_t, p_t)\big], \tag{70}$$

which replaces $v_t(m_t, n_t)$ inside the Bellman equations for v_{t-1}^{keep} and v_{t-1}^{adj}. The two frictions play complementary roles: p_t shifts the adjuster's continuous problem and therefore the level of v_t^{adj}; and ε_t^* translates the resulting gap $v_t^{\text{adj}} - v_t^{\text{keep}}$ into a smooth probability of adjusting. I derive the optimal search-effort rule, the combined envelope conditions, the Euler equations, the terminal-period solution, and the EGM algorithm for the nested model in appendix A.5.

Using a combination of endogeneous search and stochastic prices can smooth policy functions when searchfrictions are not enough. Figure 18 illustrates the S,s region for the terminal period, and figure 19 shows the MPCs in period t.

4 Conclusion

This paper introduces search as a new method for smoothing out discrete-continuous choice (DC) problems. While search can significantly reduce the size of the disconti- nuities in the decision rule of a DC problem, to get uniform smoothness around the discontinuities, we need to incorporate taste shocks as well. Together, search and taste shocks offer modelers greater control over the degree of smoothness in DC models, while rationalizing the smoothness into the agent's choice problem. With ample smoothness, discontinuities in the decision rule can be smoothed out entirely; thus, eliminating the need for incorporating an additional, computationally costly, step of identifying and removing suboptimal points generated in the EGM step for DC models. This paper studies search in the context of a consumption-saving-retirement problem; however, the framework can be easily applied to models of housing purchase, lumpy firm investments, and a wide array of other DC models. We additionally document how a coarse income-shock discretization injects phantom kinks into the decision rule, and describe a transition-matrix method that integrates the continuous income density and removes those numerical artifacts at moderate cost.

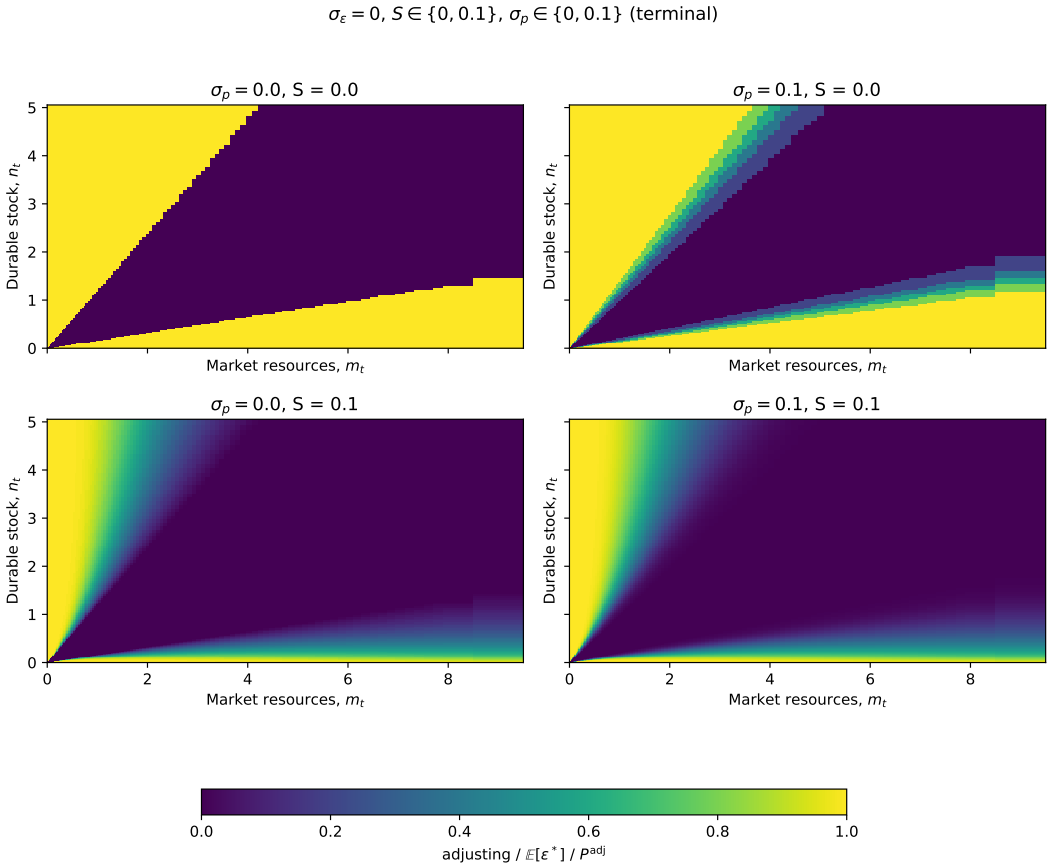
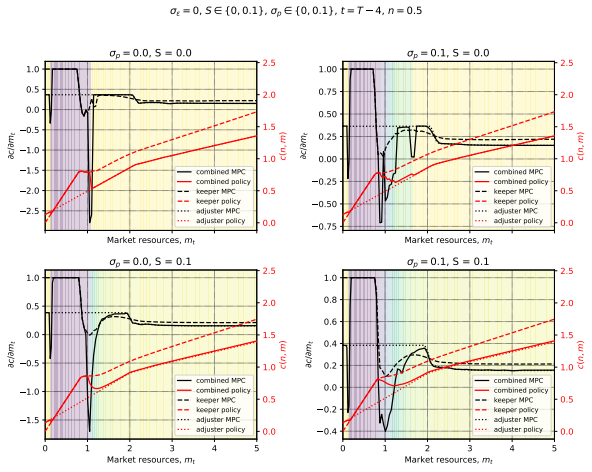
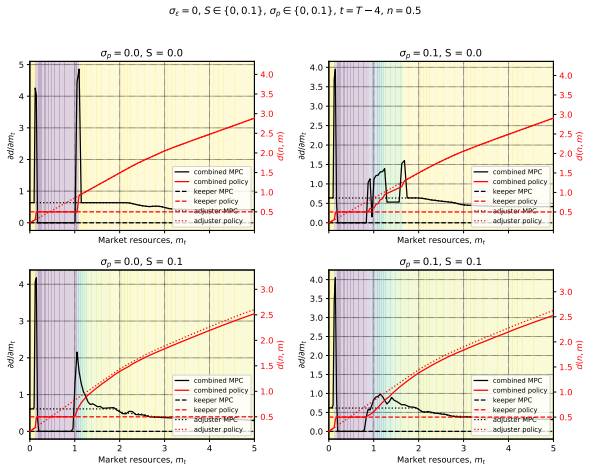


Figure 18: Adjustment regions in period T under stochastic durable prices and endogeneous search.



(a) MPC decomposition of nondurable consumption $c_t(n_t, m_t)$.



(b) MPC decomposition of durable consumption $d_t(n_t, m_t)$.

Figure 19: MPC and policy functions of non-durable and durable consumption using stochastic prices and search.

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A Model Appendix

A.1 Derive Beginning-of-Period Marginal Value Functions

Problem in Bellman Form: The value function is the hard maximum over keeping $d_t = n_t$ and adjusting $d_t \neq n_t$:

$$v_t(m_t, n_t) = \max\left\{v_t^{keep}(m_t, n_t),\ v_t^{adj}(\ell_t)\right\} \tag{71}$$

where:

$$\ell_t = m_t + (1-\tau)n_t \tag{72}$$
$$a_t = \ell_t - c_t - d_t \geq 0, \tag{73}$$
$$m_{t+1} = \mathcal{R}a_t + \theta_{t+1}, \tag{74}$$
$$n_{t+1} = (1-\delta)\, d_t \tag{75}$$
$$u(c_t, d_t) = \frac{(c_t^\alpha d_t^{1-\alpha})^{1-\rho}}{1-\rho} \tag{76}$$

Where we have the Bellman for the keeper:

$$v_t^{keep}(m_t, n_t) = \max_{c_t} u(c_t, n_t) + \beta\mathbb{E}_t\Big(v_{t+1}(m_{t+1}, n_{t+1})\Big) \tag{77}$$

s.t.

$$a_t = m_t - c_t \geq 0, \tag{78}$$
$$m_{t+1} = \mathcal{R}a_t + \theta_{t+1}, \tag{79}$$
$$n_{t+1} = (1-\delta)\, d_t \tag{80}$$
$$d_t = n_t \tag{81}$$
$$u(c_t, d_t) = \frac{(c_t^\alpha d_t^{1-\alpha})^{1-\rho}}{1-\rho} \tag{82}$$

And for the adjuster:

$$v_t^{adj}(\ell_t) = \max_{c_t, d_t} u(c_t, d_t) + \beta\mathbb{E}_t\Big(v_{t+1}(m_{t+1}, n_{t+1})\Big) \tag{83}$$

s.t.

$$\ell_t = m_t + (1-\tau)n_t \tag{84}$$
$$a_t = \ell_t - c_t - d_t \geq 0, \tag{85}$$
$$m_{t+1} = \mathcal{R}a_t + \theta_{t+1}, \tag{86}$$
$$n_{t+1} = (1-\delta)\, d_t \tag{87}$$
$$u(c_t, d_t) = \frac{(c_t^\alpha d_t^{1-\alpha})^{1-\rho}}{1-\rho} \tag{88}$$

Step 1 (chain rule / envelope): The adjuster value is a function of ℓ_t,

$$v_t^{adj} = v_t^{adj}(\ell_t), \qquad \ell_t = m_t + (1-\tau)n_t.$$

By the chain rule

$$\frac{\partial v_t^{adj}(\ell_t)}{\partial m_t} = \frac{\partial v_t^{adj}(\ell_t)}{\partial \ell_t} \cdot \frac{\partial \ell_t}{\partial m_t}, \tag{89}$$
$$\frac{\partial v_t^{adj}(\ell_t)}{\partial n_t} = \frac{\partial v_t^{adj}(\ell_t)}{\partial \ell_t} \cdot \frac{\partial \ell_t}{\partial n_t}. \tag{90}$$

Step 2 (derivatives of ℓ_t):

$$\frac{\partial \ell_t}{\partial m_t} = 1, \qquad \frac{\partial \ell_t}{\partial n_t} = 1 - \tau.$$

Step 3 (envelope for $\partial v^{adj}/\partial \ell_t$): The adjuster problem is

$$\max_{c_t, d_t} \; u(c_t, d_t) + \beta \mathbb{E}_t\big[v_{t+1}(m_{t+1}, n_{t+1})\big] \quad \text{s.t.} \quad a_t = \ell_t - c_t - d_t,$$

with the stated laws of motion. Because ℓ_t enters only through the linear resource constraint $a_t = \ell_t - c_t - d_t$, the envelope theorem implies that

$$\frac{\partial v_t^{adj}(\ell_t)}{\partial \ell_t} = \lambda_t^{adj},$$

where λ_t^{adj} is the Lagrange multiplier on the resource constraint (i.e. the marginal value of an extra unit of ℓ_t).

Step 4 (relation between λ_t^{adj} and marginal utility): The FOC w.r.t. c_t (ignoring binding additional constraints) sets

$$\lambda_t^{adj} = u_c(c_t^*, d_t^*).$$

With the specified utility,

$$u_c(c, d) = (c^\alpha d^{1-\alpha})^{-\rho}\, \alpha c^{\alpha-1} d^{1-\alpha} = \alpha\, c^{\alpha(1-\rho)-1}\, d^{(1-\alpha)(1-\rho)}.$$

If the borrowing constraint binds and has multiplier μ_t, then λ_t^{adj} equals u_c plus the relevant shadow price (i.e. $\lambda_t^{adj} = u_c +$ extra multipliers).

Step 5 (combine chain rule and envelope): Using Steps 1–4,

$$\frac{\partial v_t^{adj}}{\partial m_t} = \lambda_t^{adj} = u_c(c_t^*, d_t^*) \tag{91}$$

and

$$\frac{\partial v_t^{adj}}{\partial n_t} = (1 - \tau)\, \lambda_t^{adj} = (1 - \tau)\, u_c(c_t^*, d_t^*) \tag{92}$$

evaluated at the adjuster optimum (c_t^*, d_t^*) (and with λ_t^{adj} augmented by constraint multipliers if applicable).

Keeper: stepwise derivation (same style) In the keeper regime we have $d_t = n_t$ (fixed) and the household chooses only c_t. The state is (m_t, n_t) and

$$a_t = m_t - c_t, \qquad m_{t+1} = \mathcal{R} a_t + \theta_{t+1}, \qquad n_{t+1} = (1 - \delta)\, n_t.$$

Keeper Step 1 (envelope for m_t): Since m_t enters linearly via $a_t = m_t - c_t$, the envelope theorem gives

$$\frac{\partial v_t^{keep}(m_t, n_t)}{\partial m_t} = \lambda_t^{keep} = u_c(c_t^*, n_t) \tag{93}$$

where again λ_t^{keep} is the Lagrange multiplier on the budget and u_c is as above (add constraint multipliers if the borrowing constraint binds).

Keeper Step 2 (envelope for n_t): The durable stock n_t affects the keeper value through two channels:

1. Directly in current utility because $d_t = n_t$: marginal contribution $u_d(c_t^*, n_t)$.

2. Indirectly via $n_{t+1} = (1-\delta)\, n_t$, which affects $v_{n,t+1}$.

Applying the envelope theorem and the chain rule,

$$\frac{\partial v_t^{keep}}{\partial n_t} = u_d(c_t^*, n_t) + \beta\, \mathbb{E}_t \left[v_{n,t+1} \cdot \frac{\partial n_{t+1}}{\partial n_t} \right] \tag{94}$$

$$= u_d(c_t^*, n_t) + \beta\, \mathbb{E}_t \Big[v_{n,t+1}\, (1-\delta) \Big]. \tag{95}$$

Thus

$$\frac{\partial v_t^{keep}(m_t, n_t)}{\partial n_t} = u_d(c_t^*, n_t) + \beta(1-\delta)\, \mathbb{E}_t \big[v_{n,t+1} \big] \tag{96}$$

with

$$u_d(c, d) = (1-\alpha)\, c^{\alpha(1-\rho)}\, d^{(1-\alpha)(1-\rho)-1}.$$

Combined expression with adjust / keep indicator Let $\mathcal{I}_t$ be the indicator for *adjusting* at time t ($\mathcal{I}_t = 1$ if adjust, 0 if keep). Then the beginning-of-period partials of the value function can be summarized as

$$\frac{\partial v_t(m_t, n_t)}{\partial m_t} = \mathcal{I}_t \frac{\partial v_t^{adj}}{\partial m_t} + (1 - \mathcal{I}_t) \frac{\partial v_t^{keep}}{\partial m_t} \tag{97}$$

$$= \mathcal{I}_t \lambda_t^{adj} + (1 - \mathcal{I}_t) \lambda_t^{keep} = \frac{\partial u(c_t, d_t)}{\partial c_t} \tag{98}$$

(the final equality holds when λ's equal marginal utility of consumption; add constraint multipliers if necessary).

Similarly,

$$\frac{\partial v_t(m_t, n_t)}{\partial n_t} = \mathcal{I}_t \frac{\partial v_t^{adj}}{\partial n_t} + (1 - \mathcal{I}_t) \frac{\partial v_t^{keep}}{\partial n_t} \tag{99}$$

$$= \mathcal{I}_t (1-\tau) \lambda_t^{adj} + (1 - \mathcal{I}_t) \left\{ u_d(c_t, n_t) + \beta(1-\delta)\, \mathbb{E}_t \big[v_{n,t+1} \big] \right\}. \tag{100}$$

If $\lambda_t^{adj} = u_c(c_t^*, d_t^*)$ and $\lambda_t^{keep} = u_c(c_t^*, n_t)$, you may substitute the explicit formulas for u_c and u_d given above to obtain fully expanded expressions.

Last period This terminal-period solution serves as the initial condition for the backward induction used to find the infinite-horizon fixed point; it is not a true end-of-life period but rather a starting guess from which the backward recursion is iterated until convergence. We set the continuation value to zero, $v_{t+1} \equiv 0$. Since there is no future utility cost of debt and borrowing is prohibited ($LTV = 0$, $\underline{a} = 0$), both the keeper and the adjuster optimally set end-of-period assets to $a = 0$. For the keeper ($d_t = n_t$), this gives $c^{keep} = m_t$.

For the adjuster problem, the binding borrowing constraint is simply:

$$a = 0 \tag{101}$$

Hence, the budget constraint:

$$\ell = c + d + a = c + d \tag{102}$$

Meaning we solve:

$$\mathscr{L} = u(c,d) + \lambda\big(\ell - c - d\big) \tag{103}$$

$$\frac{\partial \mathscr{L}}{\partial c} = u_c(c,d) = \lambda \tag{104}$$

$$\frac{\partial \mathscr{L}}{\partial d} = u_d(c,d) = \lambda \tag{105}$$

Setting both first-order conditions equal and using CRRA utility $u(c,d) = \dfrac{(c^\alpha d^{1-\alpha})^{1-\rho}}{1-\rho}$:

$$u_c(c,d) = u_d(c,d) \tag{106}$$

$$\frac{\alpha(c^\alpha d^{1-\alpha})^{-\rho}}{c} = \frac{(1-\alpha)(c^\alpha d^{1-\alpha})^{-\rho}}{d} \tag{107}$$

$$\frac{c}{d} = \frac{\alpha}{1-\alpha} \tag{108}$$

$$c = d\,\frac{\alpha}{1-\alpha} \tag{109}$$

Plugging this into the budget constraint:

$$\ell = c + d = d\,\frac{\alpha}{1-\alpha} + d = d\,\frac{1}{1-\alpha} \tag{110}$$

Therefore, we have the solutions:

$$d^* = (1-\alpha)\,\ell \tag{111}$$

$$c^* = \alpha\,\ell \tag{112}$$

Note that $c^* = \alpha\ell$ is a consequence of the Cobb-Douglas aggregator, which fixes budget-share α for nondurables.

Calculate End-Of-Period Functions The standard approach is to discretize the stochastic variables with a limited amount of discrete points such as in Tauchen and others. The arising problem with a discrete choice problem is that policy functions become discontinuous and the value function is non-smooth. As a result, evaluating the next period's (marginal) value function on a few points can lead to *phantom* secondary non-concavities; that is large point masses are distributed at a few points instead of using all points in the surrounding area. The most straightforward solution is to increase the number of points with which a distribution is discretized.

Instead, I use a method of *transition matrix* or *lotteries* often used to receive model solutions[2]. That is, splitting up the transitory shock θ such that the difference between beginning of period

$$b_t = \mathcal{R}\,a_{t-1} \tag{113}$$

$$m_t - b_t = \theta_t \tag{114}$$

The algorithm is as follows:

[2]This method was developed by Matthew White for this particular purpose

1. Pre-solution step
   (a) Specify a fairly dense grid of beginning-of-period market resources and bank balances m and b
   (b) Calculate the pdf of $\theta : f_{\theta t}(m - b)$ and create a matrix with probabilities with b values by row and m values by column.
   (c) Create a Markov transition matrix by taking the row-wise sum and divide the row by it.
2. Solution step
   (a) Evaluate $v_{t+1}(m_{t+1}, n_{t+1})$ on the cross product of the dense m-grid and some grid of n.
   (b) Pre-multiply the Markov matrix by the created matrix. The resulting matrix is b row-wise and n column wise.
   (c) Construct a linear interpolation function.
   (d) Create the end-of-period (marginal) functions over the end-of-period variables a (assets) and d (durable stock) by evaluating the interim function denoted by $\mathfrak{v}(d_t, a_t)$, $\mathfrak{v}_c^{'}(d_t, a_t)$, and $\mathfrak{v}_d^{'}(d_t, a_t)$

Solve Keeper The keeper solution specifies that the beginning-of-period durable stock n_t is equal to the end-of-period durable stock d_t. Therefore, we only have to calculate optimal level of c_t and a_t which we can do by EGM and an upper-envelope algorithm to deal with non-concavities.

1. Loop over each point of the end-of-period durable stock d_t
2. Invert the end-of-period marginal value function to obtain c_t. From $u_c(c_t, n_t) = \mathfrak{v}_c^{'}(n_t, a_t)$ with $d_t = n_t$ fixed:

$$c_t = \left[\frac{\mathfrak{v}_c^{'}(n_t, a_t)}{\alpha \, n_t^{(1-\alpha)(1-\rho)}} \right]^{\frac{1}{\alpha(1-\rho)-1}}$$

3. Calculate the endogenous beginning-of-period market resources using the budget constraint

$$m_t = c_t + a_t$$

4. Use the upper envelope algorithm
5. Construct a function over n and m

Solve Adjuster Solving the adjuster problem follows the paper by Dobrescu and Shanker [2022]:

1. Loop over each end of period durable stock d_t
2. Calculate c_t over a dense grid of a_t using the EGM step. Setting the first-order condition $u_c(c_t, d_t) = \mathfrak{v}_c^{'}(d_t, a_t)$ and inverting with respect to c_t (holding d_t fixed) yields

$$c_t = \left[\frac{\mathfrak{v}_c^{'}(d_t, a_t)}{\alpha \, d_t^{(1-\alpha)(1-\rho)}} \right]^{\frac{1}{\alpha(1-\rho)-1}} \tag{115}$$

3. Find all roots of a_t which obey the first order conditions

$$0 = u_d^{'}(c, d) - \mathfrak{v}_c^{'}(d_t, a_t) + \mathfrak{v}_d^{'}(d_t, a_t) \tag{116}$$

4. Calculate ℓ_t with the budget constraint

$$\ell_t = c_t + d_t + a_t \tag{117}$$

5. Calculate the value:

$$v_t = u(c_t, d_t) + \mathfrak{v}(d_t, a_t) \tag{118}$$

6. Apply FUES

7. Create functions over (ℓ)

Discrete Choice To solve the discrete choice between adjusting and keeping, take the hard maximum of the two conditional value functions:

$$v_t(m_t, n_t) = \max \Big\{ v_t^{keep}(m_t, n_t),\ v_t^{adj}\big(m_t + (1-\tau)n_t\big) \Big\}. \tag{119}$$

Define the indicator $\mathcal{I}_t = 1$ if $v_t^{adj}(\ell_t) \geq v_t^{keep}(m_t, n_t)$ and $\mathcal{I}_t = 0$ otherwise. The combined policy functions are then

$$c_t = (1 - \mathcal{I}_t)\, c_t^{keep} + \mathcal{I}_t\, c_t^{adj}, \qquad d_t = (1 - \mathcal{I}_t)\, n_t + \mathcal{I}_t\, d_t^{adj}, \tag{120}$$

and the combined (marginal) value functions follow equations (98)–(100). When computing (marginal) value functions, note that a positive probability of an infinite conditional value implies that the combined function also takes the value infinity. To facilitate the evaluation of functions involving infinities, I store and assess their inverse representations, and subsequently invert them again when the arrays are used.

A.2 Derivations: Extreme Value Type I Taste Shocks (Section 3.4)

This subsection derives the envelope conditions, the Euler equations, and the EGM algorithm for the EV Type I taste-shock extension of Section 3.4. Throughout, $v_t^{keep}(m_t, n_t)$ and $v_t^{adj}(\ell_t)$ are exactly as defined in Section A, so the within-branch first-order conditions (43)–(44), the Step 1–5 envelope derivation for the adjuster (equations (91)–(92)), and the keeper envelope derivation (equations (93)–(96)) carry over unchanged. What changes is how the two branches are combined and, consequently, what continuation value enters the Euler equations one period back.

Log-sum inclusive value and logit probabilities Recall the EV Type I discrete choice from equations (48)–(50). The inclusive (ex-ante) value is the closed-form log-sum

$$v_t(m_t, n_t) = \sigma_\varepsilon \log \Big(\exp\big(v_t^{keep}(m_t, n_t)/\sigma_\varepsilon\big) + \exp\big(v_t^{adj}(\ell_t)/\sigma_\varepsilon\big) \Big), \tag{121}$$

with logit choice probabilities $P_t^{adj}(m_t, n_t)$ and $P_t^{keep} = 1 - P_t^{adj}$ as in (50). The taste shocks do not enter the inner maximizations defining v_t^{keep} and $v_t^{adj}(\ell_t)$, so those maximizations—and hence $c_t^{keep,*}$, $c_t^{adj,*}$, and $d_t^{adj,*}$—are independent of $(\varepsilon_t^{keep}, \varepsilon_t^{adj})$. Differentiating the log-sum (121) and

using the envelope results $\partial v_t^{keep}/\partial m_t = u_c(c_t^{keep,*}, n_t)$ and $\partial v_t^{adj}/\partial m_t = u_c(c_t^{adj,*}, d_t^{adj,*})$ from (93) and (91) yields the probability-weighted combined marginal values

$$\frac{\partial v_t(m_t, n_t)}{\partial m_t} = P_t^{keep}\, u_c(c_t^{keep,*}, n_t) + P_t^{adj}\, u_c(c_t^{adj,*}, d_t^{adj,*}), \tag{122}$$

$$\frac{\partial v_t(m_t, n_t)}{\partial n_t} = P_t^{keep}\Big[u_d(c_t^{keep,*}, n_t) + \beta(1-\delta)\,\mathbb{E}_t\big[v_{n,t+1}\big]\Big] + P_t^{adj}\,(1-\tau)\,u_c(c_t^{adj,*}, d_t^{adj,*}), \tag{123}$$

where the n-derivative of the adjuster branch uses $\partial \ell_t/\partial n_t = (1-\tau)$ as in (92). Unlike the multiplicative mean-one log-normal taste shock used in the retirement application, there is no additional scaling factor on the adjuster's marginal contribution: the EV shocks enter only through the logit weights P_t^{keep} and P_t^{adj}. As $\sigma_\varepsilon \to 0$, $P_t^{adj} \to \mathcal{I}_t$ and (122)–(123) collapse to the hard-maximum combined envelopes (98)–(100).

Expected policies The ex-ante (probability-weighted) policy functions are

$$c_t(m_t, n_t) = P_t^{keep}\, c_t^{keep,*}(m_t, n_t) + P_t^{adj}\, c_t^{adj,*}(\ell_t), \tag{124}$$

$$d_t(m_t, n_t) = P_t^{keep}\, n_t + P_t^{adj}\, d_t^{adj,*}(\ell_t), \tag{125}$$

which restates (51)–(52). Because the inclusive value (121) already integrates out the Gumbel shocks in closed form, no quadrature over taste nodes is required: v_t, $\partial v_t/\partial m_t$, $\partial v_t/\partial n_t$, c_t, and d_t are evaluated directly from the two conditional branches and the logit probabilities.

Euler equations The within-branch first-order conditions are unchanged in form, but the continuation value is now the EV-smoothed inclusive value v_{t+1} from (121) rather than the hard maximum:

$$u_c(c_t^{adj,*}, d_t^{adj,*}) \geq \beta\mathcal{R}\,\mathbb{E}_t\big[v_{m,t+1}\big], \tag{126}$$

$$u_d(c_t^{adj,*}, d_t^{adj,*}) \geq \beta\mathcal{R}\,\mathbb{E}_t\big[v_{m,t+1}\big] - \beta(1-\delta)\,\mathbb{E}_t\big[v_{n,t+1}\big], \tag{127}$$

and analogously for the keeper with $d_t = n_t$ fixed, where $v_{m,t+1} = \partial v_{t+1}/\partial m_{t+1}$ and $v_{n,t+1} = \partial v_{t+1}/\partial n_{t+1}$ are given by (122)–(123) one period ahead. Writing $\mathbb{E}_t[\cdot]$ as the expectation over θ_{t+1} and substituting (122) makes explicit that the right-hand side averages next period's income *and* the logit-weighted keep/adjust margins, e.g.

$$u_c(c_t^{adj,*}, d_t^{adj,*}) = \beta\mathcal{R}\,\mathbb{E}_{\theta_{t+1}}\Big[P_{t+1}^{keep}\, u_c(c_{t+1}^{keep,*}, n_{t+1}) + P_{t+1}^{adj}\, u_c(c_{t+1}^{adj,*}, d_{t+1}^{adj,*})\Big] \tag{128}$$

when the no-borrowing constraint does not bind.

Terminal period The within-branch terminal solutions are unaffected by the taste shocks, since $(\varepsilon_T^{keep}, \varepsilon_T^{adj})$ do not enter the inner maximizations: $c_T^{keep} = m_T$ for the keeper, and $c_T^{adj*} = \alpha\ell_T$, $d_T^{adj*} = (1-\alpha)\ell_T$ for the adjuster, exactly as derived in the "Last period" paragraph of Section A. Only the discrete resolution changes: the terminal inclusive value is the log-sum (121) at T, and the terminal expected policies follow (124)–(125).

EGM algorithm with EV Type I taste shocks No taste-shock quadrature grid is needed. In addition to the income-shock integration already used to build the end-of-period functions $\mathfrak{v}(d_t, a_t)$, $\mathfrak{v}'_c(d_t, a_t)$, and $\mathfrak{v}'_d(d_t, a_t)$ (Section A, "Calculate End-Of-Period Functions"), the algorithm proceeds as follows:

1. **Solve the within-branch problems unchanged.** Solve the keeper by EGM and upper envelope, and the adjuster by root-finding and FUES, exactly as in the "Solve Keeper" and "Solve Adjuster" paragraphs above. This yields $c_t^{keep}(m_t, n_t)$, $v_t^{keep}(m_t, n_t)$, $c_t^{adj}(\ell_t)$, $d_t^{adj}(\ell_t)$, and $v_t^{adj}(\ell_t)$.

2. **Combine with the log-sum / logit formulas on a grid of (m_t, n_t).** For each grid point, evaluate $v_t^{keep}(m_t, n_t)$ and $v_t^{adj}(\ell_t)$, form P_t^{adj} and P_t^{keep} from (50), and set

$$v_t(m_t, n_t) = \sigma_\varepsilon \log\Big(\exp\big(v_t^{keep}/\sigma_\varepsilon\big) + \exp\big(v_t^{adj}/\sigma_\varepsilon\big)\Big), \tag{129}$$

$$v_{m,t}(m_t, n_t) = P_t^{keep}\, u_c(c_t^{keep,*}, n_t) + P_t^{adj}\, u_c(c_t^{adj,*}, d_t^{adj,*}), \tag{130}$$

$$v_{n,t}(m_t, n_t) = P_t^{keep}\, v_{n,t}^{keep}(m_t, n_t) + P_t^{adj}\, (1-\tau)\, u_c(c_t^{adj,*}, d_t^{adj,*}), \tag{131}$$

   together with the expected policies (124)–(125). Numerically, evaluate the log-sum in the numerically stable form $\max\{v^{keep}, v^{adj}\} + \sigma_\varepsilon \log\big(\sum_j \exp((v^j - \max)/\sigma_\varepsilon)\big)$ (as in `calcLogSumChoiceProbs` with `smoothtype='type1'`).

3. **Feed v_t, $v_{m,t}$, $v_{n,t}$ into the period-$(t-1)$ end-of-period step,** in place of the hard-maximum objects used in the baseline model, so that the transition-matrix income integration described above operates on the EV-smoothed functions.

As $\sigma_\varepsilon \to 0$, step 2 collapses to the hard-maximum combine of the baseline algorithm. For $\sigma_\varepsilon > 0$, the inclusive value and probability-weighted policies are smooth in (m_t, n_t), which is precisely what reduces the propagation of kinks through the backward recursion.

A.3 Derivations: Search Frictions (Section 3.5)

This subsection derives the optimal search-effort rule and the smooth combined (marginal) value functions for the search extension of Section 3.5, and states the resulting EGM algorithm. As in the taste-shock extension, the within-branch Bellman problems for $v_t^{keep}(m_t, n_t)$ and $v_t^{adj}(\ell_t)$, and hence the first-order conditions (43)–(44) and the envelope results (91)–(96), are unchanged; only the way the two branches are combined, and the continuation value that results, is different.

Optimal search effort Recall the search problem from equation (54),

$$v_t(m_t, n_t) = \max_{\varepsilon_t \in [0,1]} \underbrace{\Big[(1-\varepsilon_t)\, v_t^{keep}(m_t, n_t) + \varepsilon_t\, v_t^{adj}(\ell_t) - \sigma(\varepsilon_t)\Big]}_{\equiv\, g(\varepsilon_t;\, m_t, n_t)}. \tag{132}$$

The cost function's derivative is

$$\begin{aligned}
\sigma(\varepsilon_t) &= S\big[\varepsilon_t + (1-\varepsilon_t)\log(1-\varepsilon_t)\big] \\
\sigma'(\varepsilon_t) &= S\Big[1 + \big(-\log(1-\varepsilon_t)\big) + (1-\varepsilon_t)\cdot\Big(-\frac{1}{1-\varepsilon_t}\Big)\cdot(-1)\Big] = S\big[1 - \log(1-\varepsilon_t) - 1\big] = -S\log(1-\varepsilon_t),
\end{aligned} \tag{133}$$

and $\sigma''(\varepsilon_t) = S/(1-\varepsilon_t) > 0$ for $\varepsilon_t \in [0, 1)$, so σ is strictly convex and $g(\cdot; m_t, n_t)$ is strictly concave in ε_t (linear minus convex). The first-order condition for an interior optimum is

$$\frac{\partial g}{\partial \varepsilon_t} = v_t^{adj}(\ell_t) - v_t^{keep}(m_t, n_t) - \sigma'(\varepsilon_t^*) = 0 \quad\Longleftrightarrow\quad v_t^{adj}(\ell_t) - v_t^{keep}(m_t, n_t) = -S\log(1-\varepsilon_t^*), \tag{134}$$

which, solving for ε_t^*, gives the closed-form optimal search-effort rule

$$\varepsilon_t^*(m_t, n_t) = \max\left\{0,\ 1 - \exp\left(-\frac{v_t^{adj}(\ell_t) - v_t^{keep}(m_t, n_t)}{S}\right)\right\}. \tag{135}$$

The $\max\{0, \cdot\}$ reflects the constraint $\varepsilon_t \geq 0$: when $v_t^{adj}(\ell_t) \leq v_t^{keep}(m_t, n_t)$, the unconstrained formula would return a non-positive number, but exerting effort to increase the weight on an inferior branch is never optimal, so the constraint binds at $\varepsilon_t^* = 0$. Because $\sigma'(\varepsilon_t) \to \infty$ as $\varepsilon_t \to 1^-$, the upper constraint $\varepsilon_t \leq 1$ never binds for finite $v_t^{adj} - v_t^{keep}$, consistent with search never delivering the adjustment opportunity with certainty. As $S \to 0$, $\varepsilon_t^* \to \mathbb{1}\{v_t^{adj}(\ell_t) > v_t^{keep}(m_t, n_t)\}$, the hard indicator $\mathcal{I}_t$ of the baseline model.

Envelope theorem and combined marginal value functions Because $\varepsilon_t^*(m_t, n_t)$ is chosen to satisfy the first-order (or Karush-Kuhn-Tucker) condition above, the envelope theorem implies that the indirect effect of m_t or n_t operating through ε_t^* vanishes when differentiating $v_t(m_t, n_t) = g(\varepsilon_t^*(m_t, n_t); m_t, n_t)$, leaving only the direct effects through v_t^{keep} and $v_t^{adj}(\ell_t)$:

$$\frac{\partial v_t(m_t, n_t)}{\partial m_t} = (1 - \varepsilon_t^*)\,\frac{\partial v_t^{keep}}{\partial m_t} + \varepsilon_t^*\,\frac{\partial v_t^{adj}}{\partial \ell_t} \cdot \frac{\partial \ell_t}{\partial m_t} = (1 - \varepsilon_t^*)\,\lambda_t^{keep} + \varepsilon_t^*\,\lambda_t^{adj}, \tag{136}$$

$$\frac{\partial v_t(m_t, n_t)}{\partial n_t} = (1 - \varepsilon_t^*)\,\frac{\partial v_t^{keep}}{\partial n_t} + \varepsilon_t^*\,(1 - \tau)\,\lambda_t^{adj} = (1 - \varepsilon_t^*)\Big[u_d(c_t^{keep,*}, n_t) + \beta(1 - \delta)\,\mathbb{E}_t[v_{n,t+1}]\Big] + \varepsilon_t^*\,(1 - \tau)\,\lambda_t^{adj}, \tag{137}$$

using $\lambda_t^{adj} = u_c(c_t^{adj,*}, d_t^{adj,*})$, $\lambda_t^{keep} = u_c(c_t^{keep,*}, n_t)$, and the keeper's $\partial v_t^{keep}/\partial n_t$ from (96). Equations (136)–(137) are the smooth, search-weighted analogues of the hard-indicator combined value functions (98)–(100), with ε_t^* in place of $\mathcal{I}_t$; as $S \to 0$ they coincide exactly, since $\varepsilon_t^* \to \mathcal{I}_t$.

Terminal period The within-branch terminal solutions are unaffected by search: $c_T^{keep} = m_T$ for the keeper and $c_T^{adj*} = \alpha\ell_T$, $d_T^{adj*} = (1 - \alpha)\ell_T$ for the adjuster, exactly as in the "Last period" paragraph of Section A. The terminal discrete choice instead resolves through the optimal terminal search effort $\varepsilon_T^*(m_T, n_T) = \max\{0,\ 1 - \exp(-(v_T^{adj}(\ell_T) - v_T^{keep}(m_T, n_T))/S)\}$, and the smoothed terminal value and marginal-value functions follow (54) and (136)–(137) evaluated at T; these are the objects handed to period $T - 1$.

EGM algorithm with search The algorithm follows the same outline as in the "Discrete Choice" paragraph above, with the hard maximum replaced by the smooth combination:

1. Solve the keeper (EGM plus upper envelope) and the adjuster (root-finding plus FUES) exactly as in Section A, yielding $v_t^{keep}(m_t, n_t)$, $\lambda_t^{keep}(m_t, n_t)$, $v_t^{adj}(\ell_t)$, and $\lambda_t^{adj}(\ell_t)$ on a grid of (m_t, n_t).

2. Compute the optimal search effort $\varepsilon_t^*(m_t, n_t)$ from the closed form (135) at every grid point. As in the full model, this closed form can be evaluated directly, or, for robustness when either conditional value is infinite (e.g. an infeasible branch), effort can instead be selected by a fine grid search over $[0, 1]$ that maximizes $g(\varepsilon_t; m_t, n_t)$; the two approaches coincide whenever both conditional values are finite.

3. Form the combined value function $v_t(m_t, n_t)$ from (54) and the combined marginal-value functions $v_{m,t}(m_t, n_t)$, $v_{n,t}(m_t, n_t)$ from (136)–(137).

4. Feed v_t, $v_{m,t}$, $v_{n,t}$ into the period-$(t-1)$ end-of-period step in place of the hard-maximum objects used in the baseline model.

When a conditional value is infinite (for example because a branch is infeasible), the combined value in step 3 is also infinite; as in the baseline model's discrete-choice paragraph, I store and evaluate inverse representations of the (marginal) value functions and invert them back only when the arrays are used, so that these infinities do not propagate into numerical errors during interpolation.

A.4 Derivations: Stochastic Durable Prices (Section 3.6)

This subsection derives the envelope conditions, the terminal-period solution, and the EGM algorithm for the stochastic-price extension of Section 3.6. The keeper's Bellman problem, first-order condition, and envelope results, equations (93)–(96), carry over unchanged, since the keeper never transacts in the durable market. What changes is the adjuster's budget constraint, its first-order condition with respect to d_t (equations (61)–(63) in the main text), and, consequently, how the two branches are combined.

Adjuster first-order conditions with a stochastic price The price-modified first-order conditions (61)–(63) are stated in the main text (Section 3.6). Equation (61) is identical in form to the baseline first-order condition (43): because p_t multiplies only d_t in the budget constraint, it does not appear in the c_t first-order condition, so $c_t^{adj,*}$ solves the same equation as in the baseline model *given* d_t. Equation (63) is the price-modified analogue of the baseline condition (44): the marginal cost of an extra unit of durables is now $p_t\lambda_t^{adj}$ rather than λ_t^{adj}, since each unit of d_t now costs p_t units of ℓ_t instead of one. As KKT conditions, equality in (61)–(63) holds whenever the borrowing constraint does not bind, and the left-hand side exceeds the right-hand side otherwise.

Envelope conditions Because ℓ_t still enters the adjuster's problem only through the linear resource constraint (56), and p_t is a scalar parameter rather than a function of ℓ_t, the envelope argument of Steps 1–5 above goes through unchanged with p_t held fixed:

$$\frac{\partial v_t^{adj}(\ell_t, p_t)}{\partial m_t} = \frac{\partial v_t^{adj}}{\partial \ell_t} = \lambda_t^{adj} = u_c\big(c_t^{adj,*}(p_t), d_t^{adj,*}(p_t)\big), \tag{138}$$

$$\frac{\partial v_t^{adj}(\ell_t, p_t)}{\partial n_t} = (1-\tau)\,\frac{\partial v_t^{adj}}{\partial \ell_t} = (1-\tau)\,u_c\big(c_t^{adj,*}(p_t), d_t^{adj,*}(p_t)\big), \tag{139}$$

which are formally identical to the baseline expressions (91)–(92): the price shock changes *which* $(c_t^{adj,*}, d_t^{adj,*})$ pair is optimal, but not the mapping from the marginal unit of ℓ_t to marginal utility once evaluated at that optimum. Note that p_t does not enter equation (139) directly, because the recovery value $(1-\tau)n_t$ embedded in ℓ_t is, by assumption, insulated from the current-period purchase price.

Discrete choice with a stochastic price Recall the price-contingent discrete choice from equation (66),

$$v_t(m_t, n_t, p_t) = \max\Big\{v_t^{keep}(m_t, n_t),\ v_t^{adj}(\ell_t, p_t)\Big\}. \tag{140}$$

Define the state- and price-contingent indicators $\mathbb{1}_t^{keep}(m_t, n_t, p_t)$ and $\mathbb{1}_t^{adj}(m_t, n_t, p_t) = 1 - \mathbb{1}_t^{keep}(m_t, n_t, p_t)$ analogously to the hard-maximum baseline, with $\mathbb{1}_t^{adj} = 1$ iff $v_t^{adj}(\ell_t, p_t) \geq v_t^{keep}(m_t, n_t)$. Applying

the envelope theorem to the outer maximum exactly as in equations (98)–(100), but now with the price-contingent adjuster policies $c_t^{adj,*}(p_t)$, $d_t^{adj,*}(p_t)$ from (61)–(63),

$$\frac{\partial v_t(m_t, n_t, p_t)}{\partial m_t} = \mathbb{1}_t^{keep}\, u_c(c_t^{keep,*}, n_t) + \mathbb{1}_t^{adj}\, u_c\big(c_t^{adj,*}(p_t), d_t^{adj,*}(p_t)\big), \tag{141}$$

$$\frac{\partial v_t(m_t, n_t, p_t)}{\partial n_t} = \mathbb{1}_t^{keep}\Big[u_d(c_t^{keep,*}, n_t) + \beta(1-\delta)\,\mathbb{E}_t\big[\bar{v}_{n,t+1}\big]\Big] + \mathbb{1}_t^{adj}\,(1-\tau)\,u_c\big(c_t^{adj,*}(p_t), d_t^{adj,*}(p_t)\big). \tag{142}$$

Unlike the EV Type I taste shocks of Section A.2, which reweight the two branches through logit probabilities without changing the adjuster's continuous choices, the price shock affects these marginal values only indirectly, through its effect on the optimal $(c_t^{adj,*}, d_t^{adj,*})$ pair and on the indicator functions themselves.

Integrating out the price shock As in the taste-shock case, the object that enters the time-$(t-1)$ Euler equations is the price-integrated value and marginal-value functions,

$$\bar{v}_t(m_t, n_t) \equiv \mathbb{E}_{p_t}\big[v_t(m_t, n_t, p_t)\big], \tag{143}$$

$$\bar{v}_{m,t}(m_t, n_t) \equiv \mathbb{E}_{p_t}\Big[\mathbb{1}_t^{keep}\, u_c(c_t^{keep,*}, n_t) + \mathbb{1}_t^{adj}\, u_c\big(c_t^{adj,*}(p_t), d_t^{adj,*}(p_t)\big)\Big], \tag{144}$$

$$\bar{v}_{n,t}(m_t, n_t) \equiv \mathbb{E}_{p_t}\Big[\mathbb{1}_t^{keep}\Big(u_d(c_t^{keep,*}, n_t) + \beta(1-\delta)\,\mathbb{E}_t\big[\bar{v}_{n,t+1}\big]\Big) + \mathbb{1}_t^{adj}\,(1-\tau)\,u_c\big(c_t^{adj,*}(p_t), d_t^{adj,*}(p_t)\big)\Big], \tag{145}$$

where $\bar{v}_{m,t+1}$ and $\bar{v}_{n,t+1}$ on the right-hand side of (62)–(63) one period ahead are themselves defined by the analogous expressions one period later; equation (143) restates equation (67) from the main text. As $\sigma_p \to 0$, $p_t \to 1$ a.s., $\mathbb{1}_t^{adj} \to \mathcal{I}_t$, and equations (144)–(145) collapse to the hard-maximum combined marginal-value functions (98)–(100).

The model's ex-ante (price-weighted) policy functions follow the same logic. Because the household observes p_t before choosing, the realized policy for a given draw is a hard 0/1 choice, $c_t = \mathbb{1}_t^{keep}\, c_t^{keep} + \mathbb{1}_t^{adj}\, c_t^{adj}(p_t)$ and $d_t = \mathbb{1}_t^{keep}\, n_t + \mathbb{1}_t^{adj}\, d_t^{adj}(p_t)$, while the ex-ante policy functions reported for simulation and plotting are

$$\bar{c}_t(m_t, n_t) \equiv \mathbb{E}_{p_t}\Big[\mathbb{1}_t^{keep}\, c_t^{keep} + \mathbb{1}_t^{adj}\, c_t^{adj}(p_t)\Big], \tag{146}$$

$$\bar{d}_t(m_t, n_t) \equiv \mathbb{E}_{p_t}\Big[\mathbb{1}_t^{keep}\, n_t + \mathbb{1}_t^{adj}\, d_t^{adj}(p_t)\Big]. \tag{147}$$

Euler equations The within-branch first-order conditions (61)–(63) are unchanged in form, but, since p_{t+1} is drawn afresh next period, the continuation value entering them is the price-integrated $\bar{v}_{t+1}$ rather than v_{t+1}:

$$u_c\big(c_t^{adj,*}(p_t), d_t^{adj,*}(p_t)\big) \geq \beta\mathcal{R}\,\mathbb{E}_t\big[\bar{v}_{m,t+1}\big], \tag{148}$$

$$u_d\big(c_t^{adj,*}(p_t), d_t^{adj,*}(p_t)\big) \geq p_t\,\beta\mathcal{R}\,\mathbb{E}_t\big[\bar{v}_{m,t+1}\big] - \beta(1-\delta)\,\mathbb{E}_t\big[\bar{v}_{n,t+1}\big], \tag{149}$$

and analogously for the keeper with $d_t = n_t$ fixed and no price term, since the keeper does not purchase durables. Relative to the baseline Euler equation (44), the only change is the multiplication of the discounted marginal value of assets by the current-period price p_t on the right-hand side of the durable first-order condition.

Terminal period The keeper's terminal solution is unaffected by the price shock: $c_T^{keep} = m_T$, exactly as in the "Last period" paragraph above. For the adjuster, since $v_{T+1} \equiv 0$ and borrowing is prohibited, the terminal problem is

$$\max_{c,d} \; u(c,d) \qquad \text{s.t.} \qquad \ell_T = c + p_T\, d. \tag{150}$$

The Lagrangian $\mathscr{L} = u(c,d) + \lambda(\ell_T - c - p_T d)$ gives $u_c(c,d) = \lambda$ and $u_d(c,d) = p_T\lambda$, so

$$\frac{u_d(c,d)}{u_c(c,d)} = p_T \quad \Longleftrightarrow \quad \frac{1-\alpha}{\alpha} \cdot \frac{c}{d} = p_T \quad \Longleftrightarrow \quad c = \frac{\alpha}{1-\alpha}\, p_T\, d. \tag{151}$$

Substituting into the budget constraint,

$$\ell_T = c + p_T d = p_T d\Big(\frac{\alpha}{1-\alpha} + 1\Big) = \frac{p_T\, d}{1-\alpha} \quad \Longrightarrow \quad d_T^{adj,*} = \frac{(1-\alpha)\,\ell_T}{p_T}, \qquad c_T^{adj,*} = \alpha\,\ell_T. \tag{152}$$

As in the baseline model, $c_T^{adj,*} = \alpha\ell_T$ retains the fixed Cobb-Douglas expenditure share on non-durables regardless of the price, since a proportional increase in p_T raises the cost per unit of d_T without altering the optimal *expenditure* share $(1-\alpha)\ell_T$ devoted to durables; the corresponding physical quantity purchased, $d_T^{adj,*} = (1-\alpha)\ell_T/p_T$, falls one-for-one with the price. The terminal discrete choice is $v_T(m_T, n_T, p_T) = \max\{v_T^{keep}(m_T, n_T),\, v_T^{adj}(\ell_T, p_T)\}$, and $\bar{v}_T(m_T, n_T) = \mathbb{E}_{p_T}[v_T(m_T, n_T, p_T)]$ is the continuation value handed to period $T-1$.

EGM algorithm with stochastic durable prices Let $\vec{p} = \{p_1, \ldots, p_K\}$ be quadrature nodes for the log-normal price density with weights $\vec{\mu} = \{\mu_1, \ldots, \mu_K\}$, in addition to the income-shock integration used to build the end-of-period functions $\mathfrak{v}(d_t, a_t)$, $\mathfrak{v}'_c(d_t, a_t)$, and $\mathfrak{v}'_d(d_t, a_t)$ (Section A, "Calculate End-Of-Period Functions"). Crucially, because the price shock alters the adjuster's own first-order condition (63) rather than merely reweighting an already-computed value, the adjuster problem must be re-solved at every price node, in contrast to the EV Type I taste-shock extension of Section A.2, where a single adjuster solve and a closed-form log-sum combine suffice. The algorithm proceeds as follows:

1. **Solve the keeper once,** by EGM and upper envelope exactly as in the "Solve Keeper" paragraph above; the keeper's solution does not depend on p_t.
2. **Solve the adjuster once per price node.** For each p_k, loop over the end-of-period durable grid d_t and, for each d_t, over the grid of a_t:
   (a) Invert $u_c(c_t, d_t) = \mathfrak{v}'_c(d_t, a_t)$ for c_t exactly as in the baseline model; equation (61) confirms this step does not involve p_k.
   (b) Find the roots of the price-modified first-order condition implied by (63),
   $$0 = u_d(c_t, d_t) - p_k\, \mathfrak{v}'_c(d_t, a_t) + \mathfrak{v}'_d(d_t, a_t), \tag{153}$$
   in place of $0 = u_d(c_t, d_t) - \mathfrak{v}'_c(d_t, a_t) + \mathfrak{v}'_d(d_t, a_t)$ used in the baseline model.
   (c) Compute the price-contingent liquidity $\ell_t^{(k)} = c_t + p_k\, d_t + a_t$ and the value $v_t = u(c_t, d_t) + \mathfrak{v}(d_t, a_t)$ (unchanged in form).
   (d) Apply FUES over $\ell_t^{(k)}$ to obtain $c_t^{adj}(\ell_t; p_k)$, $d_t^{adj}(\ell_t; p_k)$, and $v_t^{adj}(\ell_t; p_k)$.

3. **Combine on a grid of (m_t, n_t).** For each grid point, compute $\ell_t = m_t + (1-\tau)n_t$ (independent of p_k) and $v_t^{keep}(m_t, n_t)$, then for each node p_k evaluate $v_t^{adj}(\ell_t; p_k)$, form the indicators $\mathbb{1}^{keep}(m_t, n_t, p_k)$, $\mathbb{1}^{adj}(m_t, n_t, p_k)$, and integrate as in equations (143)–(145) (and, for reporting, equations (146)–(147)).

4. **Feed $\bar{v}_t$, $\bar{v}_{m,t}$, $\bar{v}_{n,t}$ into the period-$(t-1)$ end-of-period step,** in place of the hard-maximum v_t, $v_{m,t}$, $v_{n,t}$ used in the baseline model.

As in the baseline and taste-shock algorithms, I store and evaluate inverse representations of the (marginal) value functions so that any infinite conditional values (e.g. from an infeasible branch) do not propagate into numerical errors during interpolation.

A.5 Derivations: Search and Stochastic Prices (Section 3.7)

This subsection derives the optimal search-effort rule, the combined envelope conditions, the Euler equations, the terminal-period solution, and the EGM algorithm for the nested search-and-price environment of Section 3.7. The within-branch building blocks are taken from the standalone extensions: the keeper is exactly as in Section A; the adjuster inherits the price-contingent first-order conditions (61)–(63) and envelope results (138)–(139); and the search cost $\sigma(\varepsilon_t)$ and its derivative $\sigma'(\varepsilon_t) = -S\log(1-\varepsilon_t)$ are those of equations (53) and (133). What is new is the joint combination of search with price risk (without an additional EV Type I or multiplicative taste shock).

Adjuster first-order conditions Holding p_t fixed, the adjuster's Lagrangian is identical to the stochastic-price case,

$$\mathscr{L} = u(c_t, d_t) + \beta\mathbb{E}_t\big[\bar{v}_{t+1}(m_{t+1}, n_{t+1})\big] + \lambda_t^{adj}\big(\ell_t - c_t - p_t\, d_t - a_t\big), \tag{154}$$

where the continuation value is now the price-integrated $\bar{v}_{t+1}$ from equation (70) one period ahead (rather than the price-only integral of Section A.4, which used a hard maximum rather than search). The first-order conditions are therefore

$$\lambda_t^{adj} = u_c(c_t, d_t) = \beta\mathcal{R}\,\mathbb{E}_t\big[\bar{v}_{m,t+1}\big], \tag{155}$$

$$u_d(c_t, d_t) = p_t\,\lambda_t^{adj} - \beta(1-\delta)\,\mathbb{E}_t\big[\bar{v}_{n,t+1}\big], \tag{156}$$

with equality when the borrowing constraint does not bind. The price p_t enters only through the durable first-order condition (156), exactly as in (63). The keeper's first-order condition is likewise unchanged relative to the baseline, with $d_t = n_t$ fixed and no price term.

Optimal search effort Recall the nested discrete problem (69),

$$v_t(m_t, n_t, p_t) = \max_{\varepsilon_t\in[0,1]} \underbrace{\Big[(1-\varepsilon_t)\,v_t^{keep}(m_t, n_t) + \varepsilon_t\, v_t^{adj}(\ell_t, p_t) - \sigma(\varepsilon_t)\Big]}_{\equiv\, g(\varepsilon_t;\, m_t, n_t, p_t)}. \tag{157}$$

Because σ is strictly convex, g is strictly concave in ε_t. The interior first-order condition is

$$v_t^{adj}(\ell_t, p_t) - v_t^{keep}(m_t, n_t) - \sigma'(\varepsilon_t^*) = 0 \quad\Longleftrightarrow\quad v_t^{adj}(\ell_t, p_t) - v_t^{keep}(m_t, n_t) = -S\log(1-\varepsilon_t^*), \tag{158}$$

which yields the closed-form, state- and price-contingent optimal search effort

$$\varepsilon_t^*(m_t, n_t, p_t) = \max\left\{0,\ 1 - \exp\left(-\frac{v_t^{adj}(\ell_t, p_t) - v_t^{keep}(m_t, n_t)}{S}\right)\right\}. \tag{159}$$

Relative to the search-only rule (135), the gap that drives effort is now $v_t^{adj}(\ell_t, p_t) - v_t^{keep}$: a low durable price (which raises v_t^{adj}) increases ε_t^*, while a high price reduces it. As in the search-only case, $\varepsilon_t^* = 0$ whenever $v_t^{adj} \le v_t^{keep}$, and the upper constraint $\varepsilon_t \le 1$ never binds for finite S. Special cases recover the earlier extensions: with $p_t \equiv 1$, equation (159) reduces to (135); with $S \to 0$, $\varepsilon_t^* \to \mathbb{1}\{v_t^{adj}(\ell_t, p_t) > v_t^{keep}\}$, the hard indicator of the price-contingent hard maximum.

Envelope conditions and combined marginal values Because ε_t^* is chosen optimally, the envelope theorem eliminates indirect effects through search effort. Differentiating $v_t = g(\varepsilon_t^*; m_t, n_t, p_t)$ and using the adjuster envelopes (138)–(139) together with the keeper envelopes (93)–(96) gives

$$\frac{\partial v_t(m_t, n_t, p_t)}{\partial m_t} = (1 - \varepsilon_t^*)\, u_c(c_t^{keep,*}, n_t) + \varepsilon_t^*\, u_c\big(c_t^{adj,*}(p_t), d_t^{adj,*}(p_t)\big), \tag{160}$$

$$\frac{\partial v_t(m_t, n_t, p_t)}{\partial n_t} = (1 - \varepsilon_t^*)\Big[u_d(c_t^{keep,*}, n_t) + \beta(1-\delta)\,\mathbb{E}_t\big[\bar{v}_{n,t+1}\big]\Big] + \varepsilon_t^*\,(1-\tau)\,u_c\big(c_t^{adj,*}(p_t), d_t^{adj,*}(p_t)\big). \tag{161}$$

Equations (160)–(161) nest the earlier combined marginal-value formulas: with $\varepsilon_t^* \in \{0, 1\}$ they collapse to the price-only indicators (141)–(142); with $p_t \equiv 1$ they collapse to the search-only envelopes (136)–(137). The price p_t enters only indirectly through $(c_t^{adj,*}, d_t^{adj,*})$ and through ε_t^* itself.

Integrating out price shocks The objects that enter the time-$(t-1)$ Euler equations are the price-integrated value and marginal-value functions

$$\bar{v}_t(m_t, n_t) \equiv \mathbb{E}_{p_t}\big[v_t(m_t, n_t, p_t)\big], \tag{162}$$

$$\bar{v}_{m,t}(m_t, n_t) \equiv \mathbb{E}_{p_t}\Big[(1 - \varepsilon_t^*)\, u_c(c_t^{keep,*}, n_t) + \varepsilon_t^*\, u_c\big(c_t^{adj,*}(p_t), d_t^{adj,*}(p_t)\big)\Big], \tag{163}$$

$$\bar{v}_{n,t}(m_t, n_t) \equiv \mathbb{E}_{p_t}\Big[(1 - \varepsilon_t^*)\Big(u_d(c_t^{keep,*}, n_t) + \beta(1-\delta)\,\mathbb{E}_t\big[\bar{v}_{n,t+1}\big]\Big) + \varepsilon_t^*\,(1-\tau)\,u_c\big(c_t^{adj,*}(p_t), d_t^{adj,*}(p_t)\big)\Big], \tag{164}$$

where $\varepsilon_t^* = \varepsilon_t^*(m_t, n_t, p_t)$ from (159), and equation (162) restates (70) from the main text. The model's ex-ante (price-weighted) policy functions are

$$\bar{c}_t(m_t, n_t) \equiv \mathbb{E}_{p_t}\Big[(1 - \varepsilon_t^*)\, c_t^{keep} + \varepsilon_t^*\, c_t^{adj}(p_t)\Big], \tag{165}$$

$$\bar{d}_t(m_t, n_t) \equiv \mathbb{E}_{p_t}\Big[(1 - \varepsilon_t^*)\, n_t + \varepsilon_t^*\, d_t^{adj}(p_t)\Big], \tag{166}$$

$$\bar{\varepsilon}_t(m_t, n_t) \equiv \mathbb{E}_{p_t}\big[\varepsilon_t^*(m_t, n_t, p_t)\big]. \tag{167}$$

Because the household observes p_t before choosing ε_t, the *realized* policy for a given draw is the search-weighted mix of keep and adjust at that price; equations (165)–(167) are the objects reported for simulation and plotting.

Euler equations Combining (155)–(156) with the integrated continuation values (163)–(164) yields the within-branch Euler equations

$$u_c\big(c_t^{adj,*}(p_t), d_t^{adj,*}(p_t)\big) \geq \beta\mathcal{R}\,\mathbb{E}_t\big[\bar{v}_{m,t+1}\big], \tag{168}$$

$$u_d\big(c_t^{adj,*}(p_t), d_t^{adj,*}(p_t)\big) \geq p_t\,\beta\mathcal{R}\,\mathbb{E}_t\big[\bar{v}_{m,t+1}\big] - \beta(1-\delta)\,\mathbb{E}_t\big[\bar{v}_{n,t+1}\big], \tag{169}$$

and analogously for the keeper with $d_t = n_t$ fixed and no price term. Relative to the baseline, the only structural change inside the adjuster's durable Euler equation is the factor p_t on the opportunity-cost term; relative to the price-only extension, the continuation values $\bar{v}_{m,t+1}$ and $\bar{v}_{n,t+1}$ additionally average over next period's search effort.

Terminal period The within-branch terminal solutions are those of the price extension: $c_T^{keep} = m_T$ for the keeper, and

$$c_T^{adj,*} = \alpha\,\ell_T, \qquad d_T^{adj,*} = \frac{(1-\alpha)\,\ell_T}{p_T}, \tag{170}$$

from equation (152). The terminal discrete problem then resolves through the optimal terminal search effort

$$\varepsilon_T^*(m_T, n_T, p_T) = \max\left\{0,\ 1 - \exp\left(-\frac{v_T^{adj}(\ell_T, p_T) - v_T^{keep}(m_T, n_T)}{S}\right)\right\}, \tag{171}$$

the terminal value follows (69) evaluated at T, and $\bar{v}_T(m_T, n_T) = \mathbb{E}_{p_T}[v_T(m_T, n_T, p_T)]$ is the continuation value handed to period $T-1$.

EGM algorithm with search and stochastic prices Let $\vec{p} = \{p_1, \ldots, p_{K_p}\}$ be quadrature nodes for the log-normal price density with weights $\vec{\mu}^p$, in addition to the income-shock integration used to build $\mathfrak{v}(d_t, a_t)$, $\mathfrak{v}'_c(d_t, a_t)$, and $\mathfrak{v}'_d(d_t, a_t)$. The algorithm nests the standalone price and search steps:

1. **Solve the keeper once,** by EGM and upper envelope exactly as in the "Solve Keeper" paragraph of Section A; the keeper does not depend on p_t.

2. **Solve the adjuster once per price node.** For each p_k, proceed exactly as in the price-only algorithm of Section A.4: invert $u_c = \mathfrak{v}'_c$ for c_t, find roots of

$$0 = u_d(c_t, d_t) - p_k\,\mathfrak{v}'_c(d_t, a_t) + \mathfrak{v}'_d(d_t, a_t), \tag{172}$$

   form $\ell_t^{(k)} = c_t + p_k\,d_t + a_t$, and apply FUES to obtain $c_t^{adj}(\ell_t; p_k)$, $d_t^{adj}(\ell_t; p_k)$, and $v_t^{adj}(\ell_t; p_k)$.

3. **Combine on a grid of (m_t, n_t), integrating over p_k.** For each grid point, compute $\ell_t = m_t + (1-\tau)n_t$ and $v_t^{keep}(m_t, n_t)$. Then for each p_k:

   (a) Evaluate $v_t^{adj}(\ell_t; p_k)$ and form the optimal search effort $\varepsilon^*(m_t, n_t, p_k)$ from (159) (or, for robustness when a conditional value is infinite, by a fine grid search over $[0,1]$ maximizing g).

   (b) Form the price-contingent value and marginal values from (69) and (160)–(161).

   Integrate with weights μ_k^p to obtain $\bar{v}_t$, $\bar{v}_{m,t}$, $\bar{v}_{n,t}$ as in (162)–(164) (and, for reporting, $\bar{c}_t$, $\bar{d}_t$, $\bar{\varepsilon}_t$ from (165)–(167)).

4. **Feed $\bar{v}_t$, $\bar{v}_{m,t}$, $\bar{v}_{n,t}$ into the period-$(t-1)$ end-of-period step,** in place of the hard-maximum objects used in the baseline model.

As in the earlier algorithms, I store and evaluate inverse representations of the (marginal) value functions so that infinite conditional values do not propagate into numerical errors during interpolation. Computational cost scales with K_p for both the adjuster solve and the combine step.